

last time

register renaming motivation

- different versions of registers

- out-of-order means multiple active versions

out-of-order pipeline

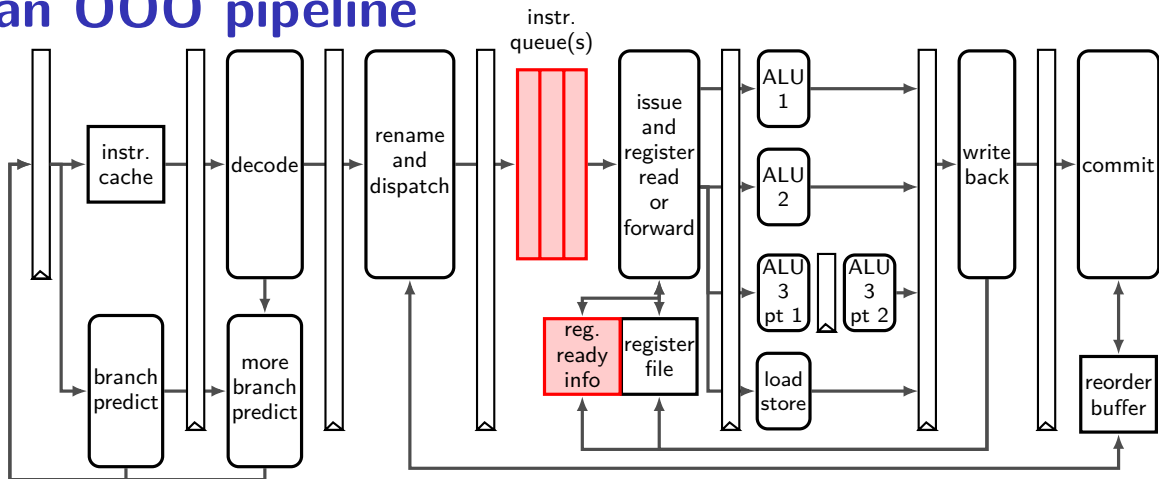
- in-order: fetch (w/ branch predict) + decode + rename + dispatch

- out-of-order: issue + execute + writeback

- in-order: commit

register renaming mechanics

an OOO pipeline



instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc

... ..

execution unit

ALU 1

ALU 2

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending
%x07	pending
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

...

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc

... ..

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending
%x07	pending
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit *cycle# 1*

ALU 1 **1**

ALU 2

...

instruction queue and dispatch

instruction queue

#	instruction
1	<code>addq %x01, %x05 → %x06</code>
2	<code>addq %x02, %x06 → %x07</code>
3	<code>addq %x03, %x07 → %x08</code>
4	<code>cmpq %x04, %x08 → %x09.cc</code>
5	<code>jne %x09.cc, ...</code>
6	<code>addq %x01, %x08 → %x10</code>
7	<code>addq %x02, %x10 → %x11</code>
8	<code>addq %x03, %x11 → %x12</code>
9	<code>cmpq %x04, %x12 → %x13.cc</code>

... ..

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending
%x07	pending
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit cycle# 1

ALU 1 1

ALU 2

...

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc

... ..

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit cycle# 1
 ALU 1 1
 ALU 2 —

...

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit	cycle#	1	2
ALU 1		1	2
ALU 2		—	—

...

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending ready
%x08	pending ready
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit	cycle#	1	2	3	...
ALU 1		1	2	3	
ALU 2		—	—	—	

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending ready
%x08	pending ready
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit	cycle#	1	2	3	...
ALU 1		1	2	3	
ALU 2		—	—	—	

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending ready
%x08	pending ready
%x09	pending ready
%x10	pending ready
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit	cycle#	1	2	3	4	...
ALU 1		1	2	3	4	
ALU 2		—	—	—	6	

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending ready
%x08	pending ready
%x09	pending ready
%x10	pending ready
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit	cycle#	1	2	3	4	...
ALU 1		1	2	3	4	
ALU 2		—	—	—	6	

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending ready
%x08	pending ready
%x09	pending ready
%x10	pending ready
%x11	pending ready
%x12	pending
%x13	pending
...	...

execution unit	cycle#	1	2	3	4	5	...
ALU 1		1	2	3	4	5	
ALU 2		—	—	—	6	7	

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending ready
%x08	pending ready
%x09	pending ready
%x10	pending ready
%x11	pending ready
%x12	pending
%x13	pending
...	...

execution unit	cycle#	1	2	3	4	5	6	...
ALU 1		1	2	3	4	5	8	
ALU 2		—	—	—	6	7	—	

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending ready
%x08	pending ready
%x09	pending ready
%x10	pending ready
%x11	pending ready
%x12	pending ready
%x13	pending
...	...

execution unit	cycle#	1	2	3	4	5	6	7	...
ALU 1		1	2	3	4	5	8	9	
ALU 2		—	—	—	6	7	—	...	

instruction queue and dispatch

instruction queue

#	instruction
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc

... ..

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	pending ready
%x08	pending ready
%x09	pending ready
%x10	pending ready
%x11	pending ready
%x12	pending ready
%x13	pending ready
...	...

execution unit	cycle#	1	2	3	4	5	6	7	...
ALU 1		1	2	3	4	5	8	9	
ALU 2		—	—	—	6	7	—	...	

instruction queue and dispatch

instruction queue

#	instruction
1	movq (%x04) → %x06
2	movq (%x05) → %x07
3	addq %x01, %x02 → %x08
4	addq %x01, %x06 → %x09
5	addq %x01, %x07 → %x10

... ..

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	
%x07	
%x08	
%x09	
%x10	
...	...

execution unit cycle# 1 2 3 4 5 6 7 ...

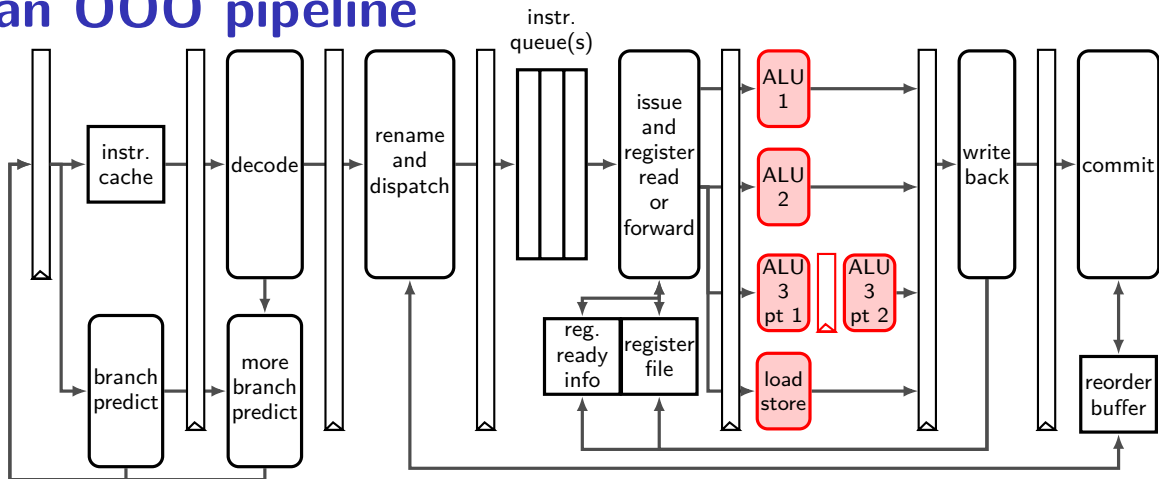
ALU

data cache

↑
assume

1 cycle/access

an OOO pipeline



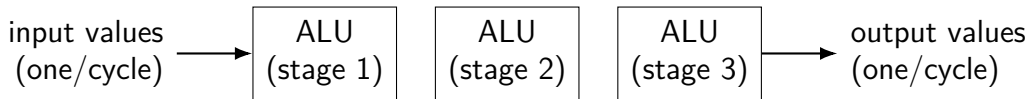
execution units AKA functional units (1)

where actual work of instruction is done

e.g. the actual ALU, or data cache

sometimes pipelined:

(here: 1 op/cycle; 3 cycle latency)



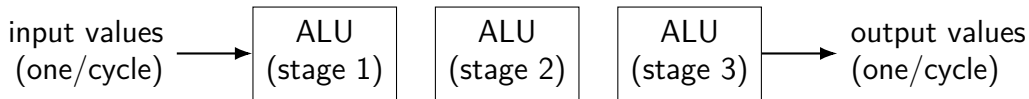
execution units AKA functional units (1)

where actual work of instruction is done

e.g. the actual ALU, or data cache

sometimes pipelined:

(here: 1 op/cycle; 3 cycle latency)



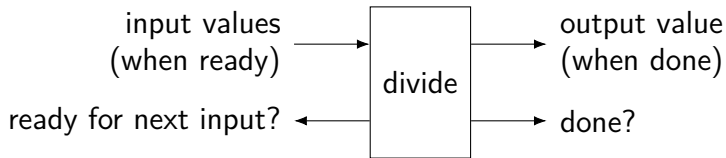
exercise: how long to compute $A \times (B \times (C \times D))$?

execution units AKA functional units (2)

where actual work of instruction is done

e.g. the actual ALU, or data cache

sometimes unpipelined:



instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit

ALU 1 (add, cmp, jxx)

ALU 2 (add, cmp, jxx)

ALU 3 (mul) start

ALU 3 (mul) end

reg	status
%x01	ready
%x02	ready
%x03	pending
%x04	ready
%x05	ready
%x06	pending
%x07	ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
%x14	pending
...	

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit

ALU 1 (add, cmp, jxx)

ALU 2 (add, cmp, jxx)

ALU 3 (mul) start

ALU 3 (mul) end

reg	status
%x01	ready
%x02	ready
%x03	pending
%x04	ready
%x05	ready
%x06	pending
%x07	ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
%x14	pending
...	

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit	cycle#
ALU 1 (add, cmp, jxx)	1
ALU 2 (add, cmp, jxx)	—
ALU 3 (mul) start	2
ALU 3 (mul) end	2

reg	status
%x01	ready
%x02	ready
%x03	pending
%x04	ready
%x05	ready
%x06	pending
%x07	ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
%x14	pending
...	

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit	cycle#	1	2
ALU 1 (add, cmp, jxx)	1	6	
ALU 2 (add, cmp, jxx)	—	—	
ALU 3 (mul) start	2	3	
ALU 3 (mul) end		2	3

reg	status
%x01	ready
%x02	ready
%x03	pending ready
%x04	ready
%x05	ready
%x06	pending (still)
%x07	ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
%x14	pending
...	

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit	cycle#	1	2	3
ALU 1 (add, cmp, jxx)	1	6	—	—
ALU 2 (add, cmp, jxx)	—	—	—	—
ALU 3 (mul) start	2	3	7	—
ALU 3 (mul) end		2	3	7

reg	status
%x01	ready
%x02	ready
%x03	pending ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	ready
%x08	pending (still)
%x09	pending
%x10	pending
%x11	pending ready
%x12	pending
%x13	pending
%x14	pending
...	

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit	cycle#	1	2	3	4
ALU 1 (add, cmp, jxx)	1	6	—	—	4
ALU 2 (add, cmp, jxx)	—	—	—	—	—
ALU 3 (mul) start	2	3	7	8	
ALU 3 (mul) end		2	3	7	8

reg	status
%x01	ready
%x02	ready
%x03	pending ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	ready
%x08	pending ready
%x09	pending ready
%x10	pending
%x11	pending ready
%x12	pending (still)
%x13	pending
%x14	pending
...	

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit	cycle#	1	2	3	4	5
ALU 1 (add, cmp, jxx)	1	6	—	4	5	5
ALU 2 (add, cmp, jxx)	—	—	—	—	—	—
ALU 3 (mul) start	2	3	7	8	—	—
ALU 3 (mul) end		2	3	7	8	

reg	status
%x01	ready
%x02	ready
%x03	pending ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	ready
%x08	pending ready
%x09	pending ready
%x10	pending
%x11	pending ready
%x12	pending ready
%x13	pending (still)
%x14	pending
...	

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit	cycle#	1	2	3	4	5
ALU 1 (add, cmp, jxx)	1	6	—	4	5	
ALU 2 (add, cmp, jxx)	—	—	—	—	—	
ALU 3 (mul) start	2	3	7	8	—	
ALU 3 (mul) end		2	3	7	8	

reg	status
%x01	ready
%x02	ready
%x03	pending ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	ready
%x08	pending ready
%x09	pending ready
%x10	pending
%x11	pending ready
%x12	pending ready
%x13	pending ready
%x14	pending
...	

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit	cycle#	1	2	3	4	5
ALU 1 (add, cmp, jxx)		1	6	—	4	5
ALU 2 (add, cmp, jxx)		—	—	—	—	—
ALU 3 (mul) start		2	3	7	8	—
ALU 3 (mul) end			2	3	7	8

reg	status
%x01	ready
%x02	ready
%x03	pending ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	ready
%x08	pending ready
%x09	pending ready
%x10	pending
%x11	pending ready
%x12	pending ready
%x13	pending ready
%x14	pending ready
...	...

6
9
—

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	instruction
1	add %x01, %x02 → %x03
2	imul %x04, %x05 → %x06
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...

... ..

execution unit	cycle#	1	2	3	4	5
ALU 1 (add, cmp, jxx)		1	6	—	4	5
ALU 2 (add, cmp, jxx)		—	—	—	—	—
ALU 3 (mul) start		2	3	7	8	—
ALU 3 (mul) end			2	3	7	8

reg	status
%x01	ready
%x02	ready
%x03	pending ready
%x04	ready
%x05	ready
%x06	pending ready
%x07	ready
%x08	pending ready
%x09	pending ready
%x10	pending
%x11	pending ready
%x12	pending ready
%x13	pending ready
%x14	pending ready
6	7... ..

9 10
— —

register renaming: missing pieces

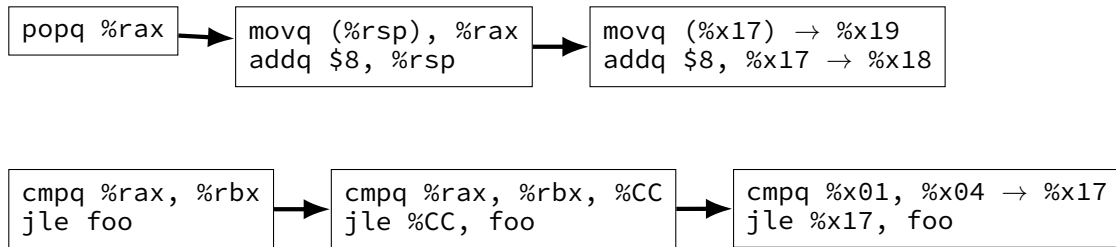
what about “hidden” inputs like `%rsp`, condition codes?

one solution: translate to instructions with additional register parameters

- making `%rsp` explicit parameter

- turning hidden condition codes into operands!

bonus: can also translate complex instructions to simpler ones



OOO limitations

can't always find instructions to run

- plenty of instructions, but all depend on unfinished ones

- programmer can adjust program to help this

need to track all uncommitted instructions

- can only go so far ahead

- e.g. Intel Skylake: 224-entry reorder buffer, 168 physical registers

branch misprediction has a big cost (relative to pipelined)

- e.g. Intel Skylake: up to approx. 16 cycles (v. 2 for simple pipelined CPU)

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branch misprediction has a big cost (relative to pipelined)

e.g. Intel Skylake: up to approx. 16 cycles (v. 2 for simple pipelined CPU)

some performance examples

```
example1:  
    movq $1000000000000, %rax  
loop1:  
    addq %rbx, %rcx  
    decq %rax  
    jge loop1  
    ret
```

about 30B instructions
my desktop: approx 2.65 sec

```
example2:  
    movq $1000000000000, %rax  
loop2:  
    addq %rbx, %rcx  
    addq %r8, %r9  
    decq %rax  
    jge loop2  
    ret
```

about 40B instructions
my desktop: approx 2.65 sec

some performance examples

```
example1:  
    movq $1000000000000, %rax  
loop1:  
    addq %rbx, %rcx  
    decq %rax  
    jge loop1  
    ret
```

about 30B instructions

my desktop: approx 2.65 sec

```
example2:  
    movq $1000000000000, %rax  
loop2:  
    addq %rbx, %rcx  
    addq %r8, %r9  
    decq %rax  
    jge loop2  
    ret
```

about 40B instructions

my desktop: approx 2.65 sec

importance of prediction

5-stage pipeline, predict not-taken versus always stall
hypothetical instruction mix

		cycles	cycles
taken jXX	3%	(predict)	(stall)
non-taken jXX	5%	not-taken)	no predict)
others	92%	1*	1*

importance of prediction

5-stage pipeline, predict not-taken versus always stall
hypothetical instruction mix

		cycles	cycles
taken jXX	3%	(predict)	(stall)
non-taken jXX	5%	not-taken)	no predict)
others	92%	1*	1*

prediction and OOO

deeper pipeline — much higher misprediction penalty

static branch prediction

forward (target > PC) not taken; backward taken

intuition: loops:

```
LOOP: ...
```

```
    ...
```

```
    je LOOP
```

```
LOOP: ...
```

```
    jne SKIP_LOOP
```

```
    ...
```

```
    jmp LOOP
```

```
SKIP_LOOP:
```


predict: repeat last

PC of branch

0x40042A

hash function

index *prediction/
last result?*

0

taken (1)

1

not taken (0)

2

taken (1)

3

taken (1)

...

...

14

not taken (0)

15

taken (1)

predict: repeat last

PC of branch

0x40042A

hash function

index *prediction/
last result?*

0 taken (1)

1 not taken (0)

2 taken (1)

3 taken (1)

...

14 not taken (0)

typical choice: some bits of branch address
for our example: will use bits 4-7

predict: repeat last

PC of branch

0x40042A

hash function

<i>index</i>	<i>prediction/ last result?</i>
0	taken (1)
1	not taken (0)
2	taken (1)
3	taken (1)
...	...
14	not taken (0)
15	taken (1)

predict: repeat last

PC of branch

0x40042A

hash function

<i>index</i>	<i>prediction/ last result?</i>
0	taken (1)
1	not taken (0)
2	taken (1)
3	taken (1)
...	...
14	not taken (0)
15	taken (1)

prediction
to fetch stage

predict: repeat last

PC of branch

0x40042A

hash function

index prediction/
last result?

0

taken (1)

1

not taken (0)

2

taken (1)

3

taken (1)

...

...

14

not taken (0)

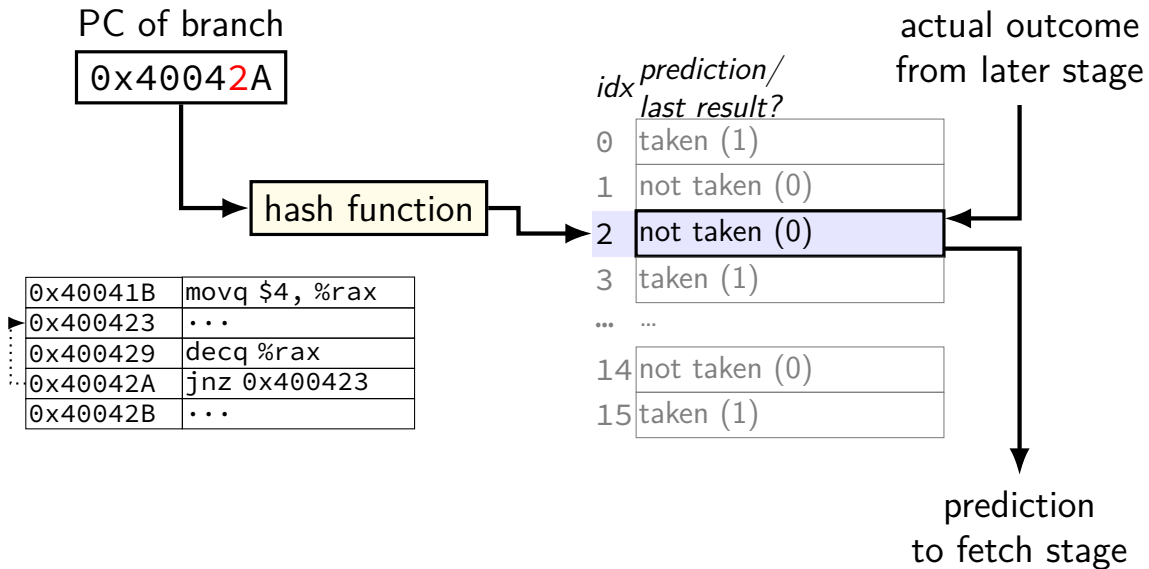
15

taken (1)

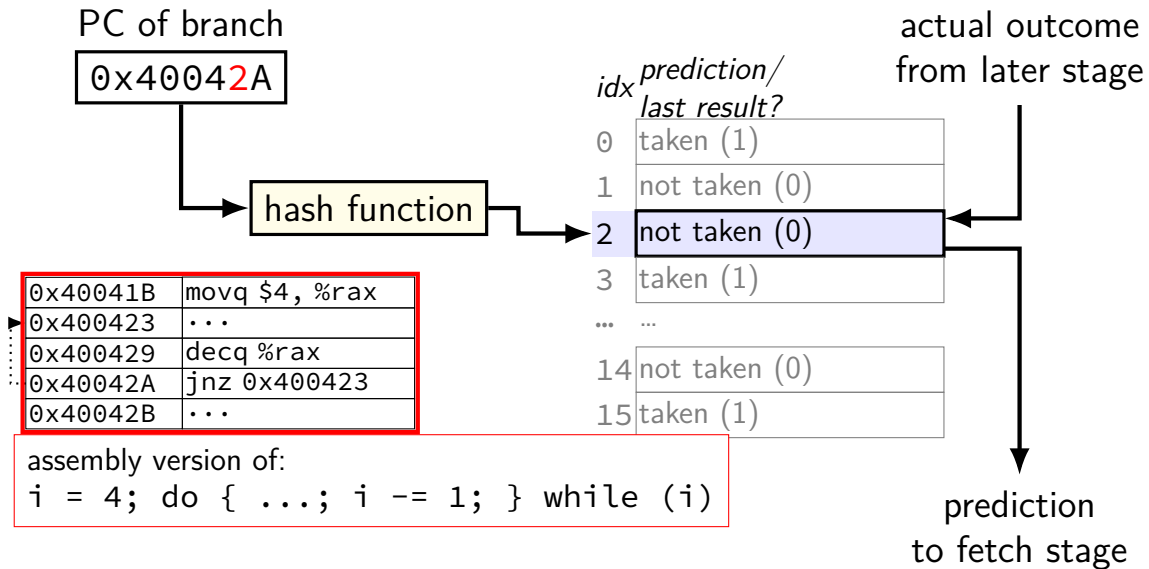
actual outcome
(from later stage)

prediction
to fetch stage

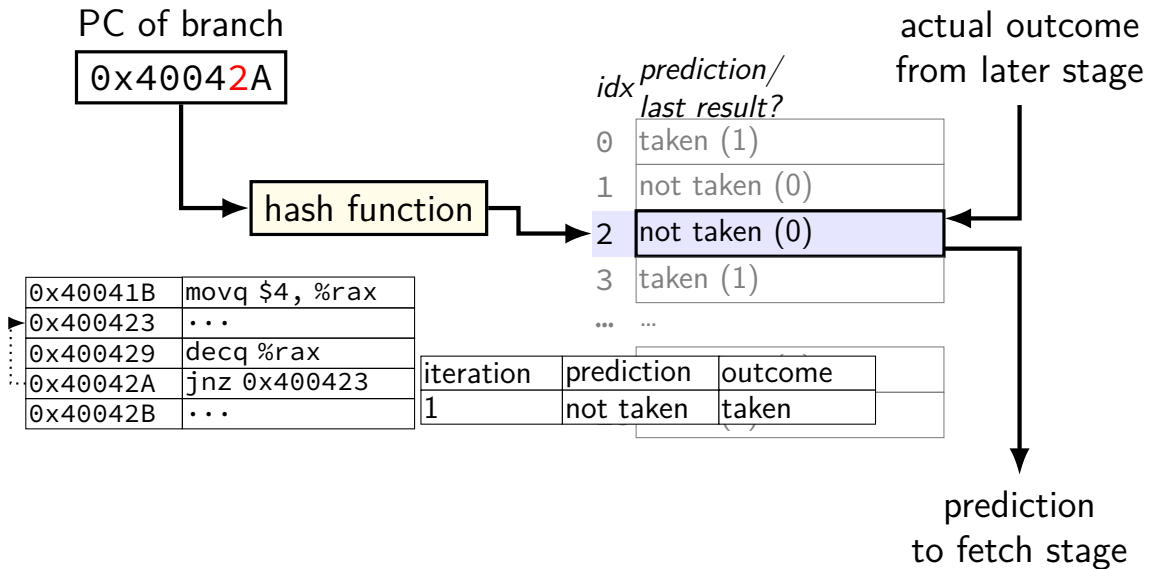
example



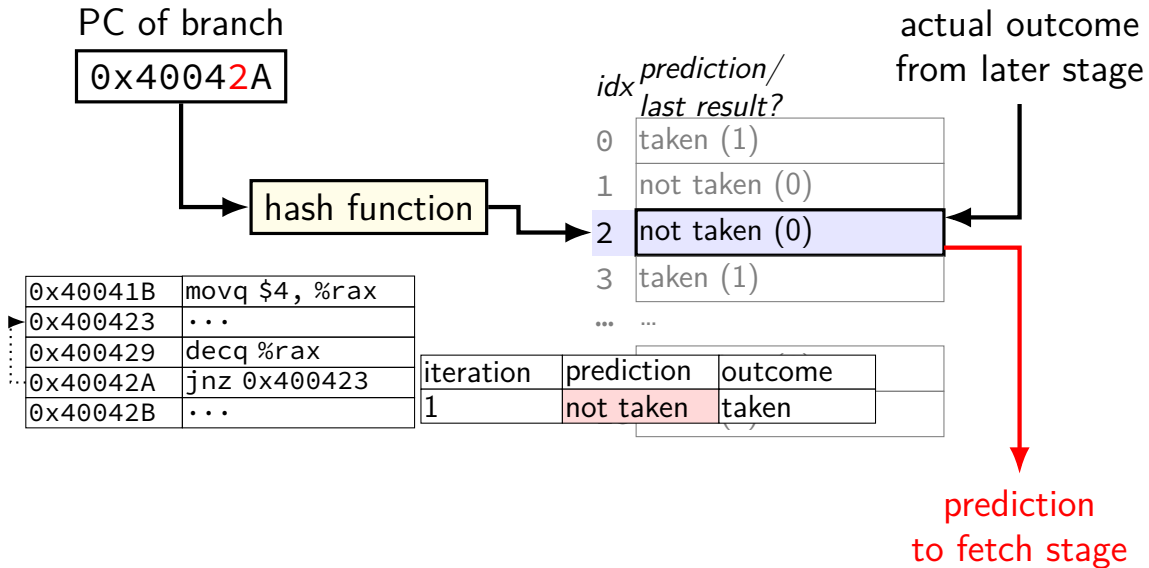
example



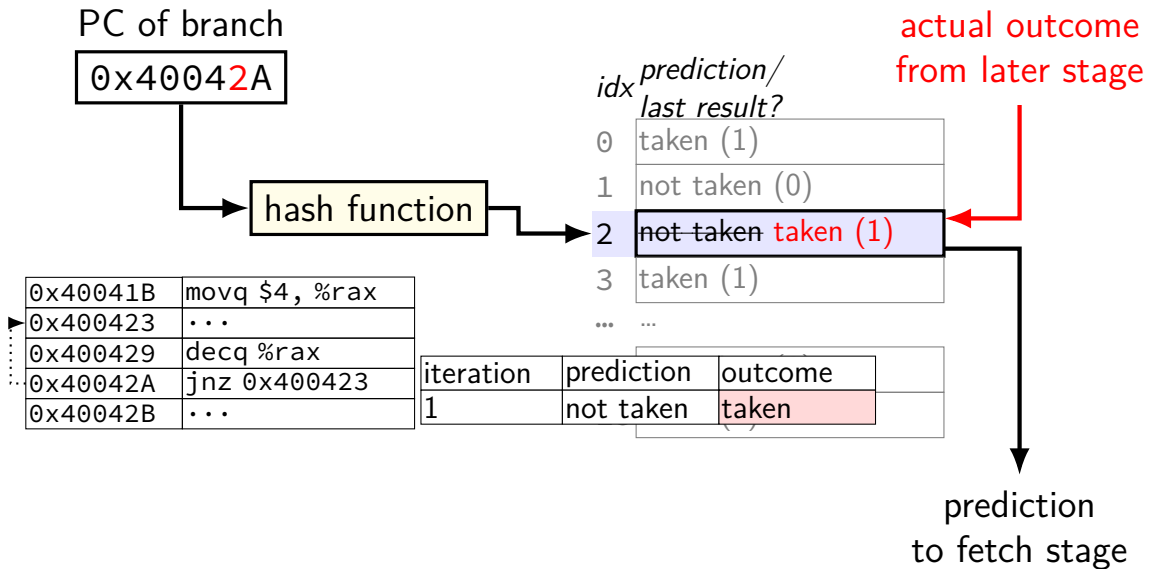
example



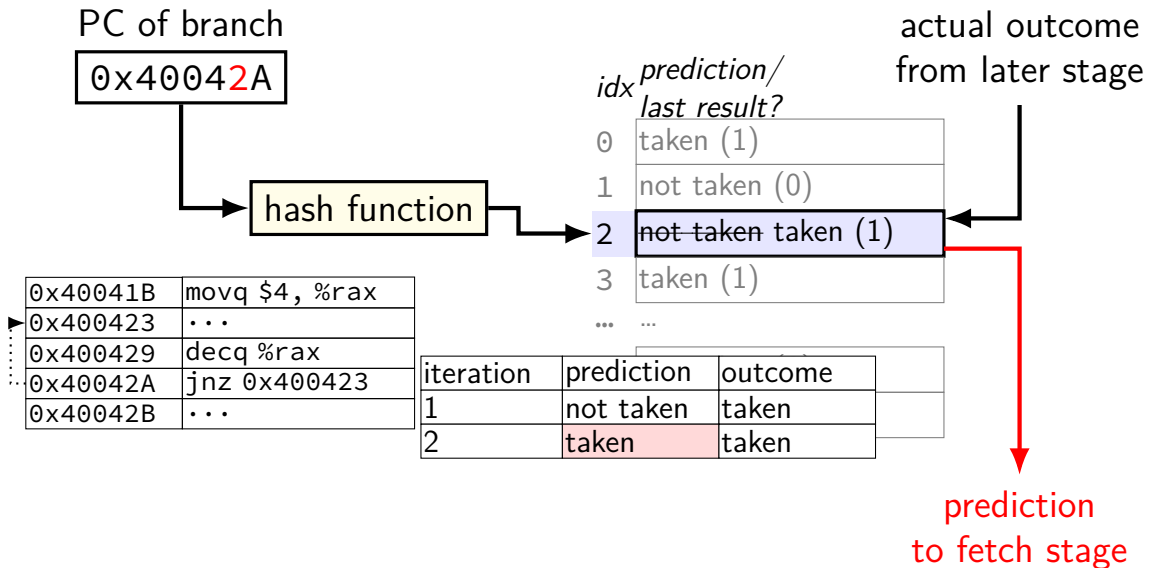
example



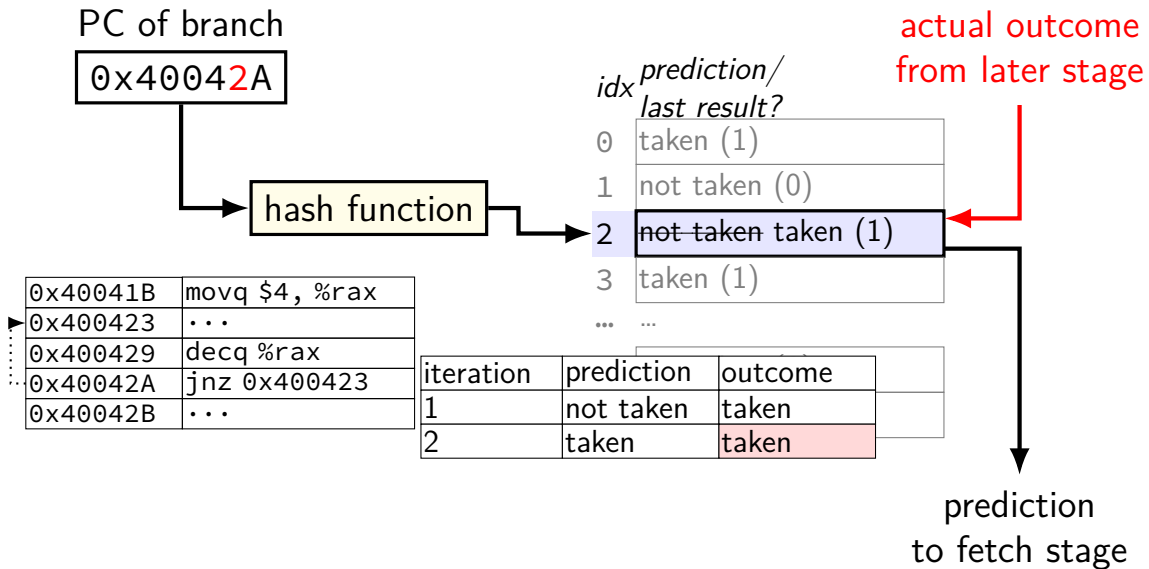
example



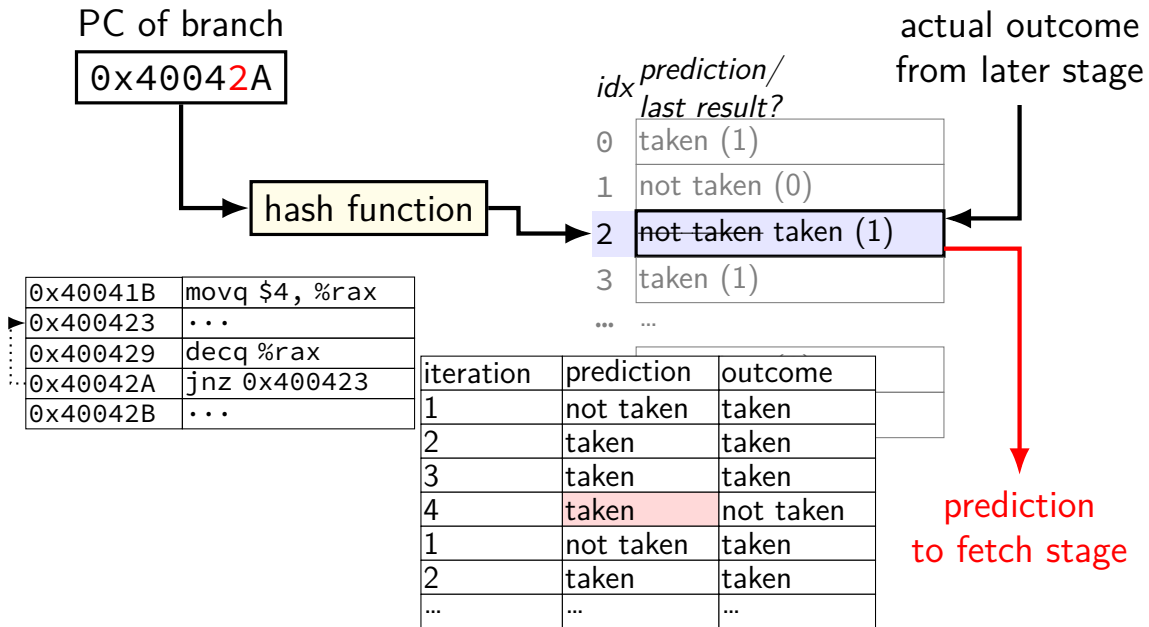
example



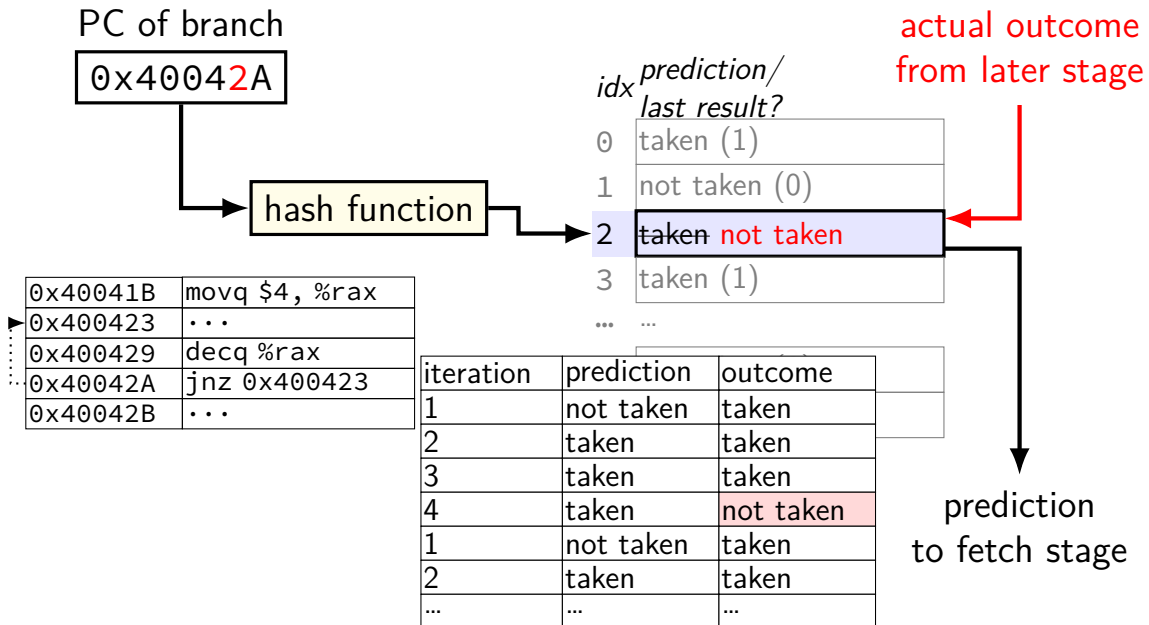
example



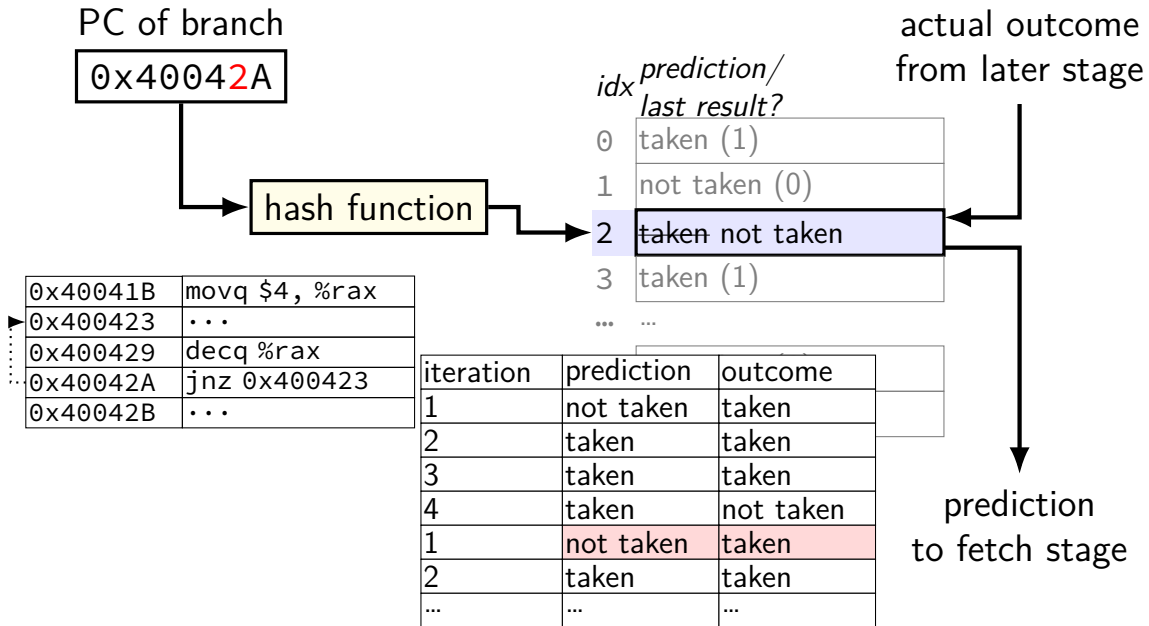
example



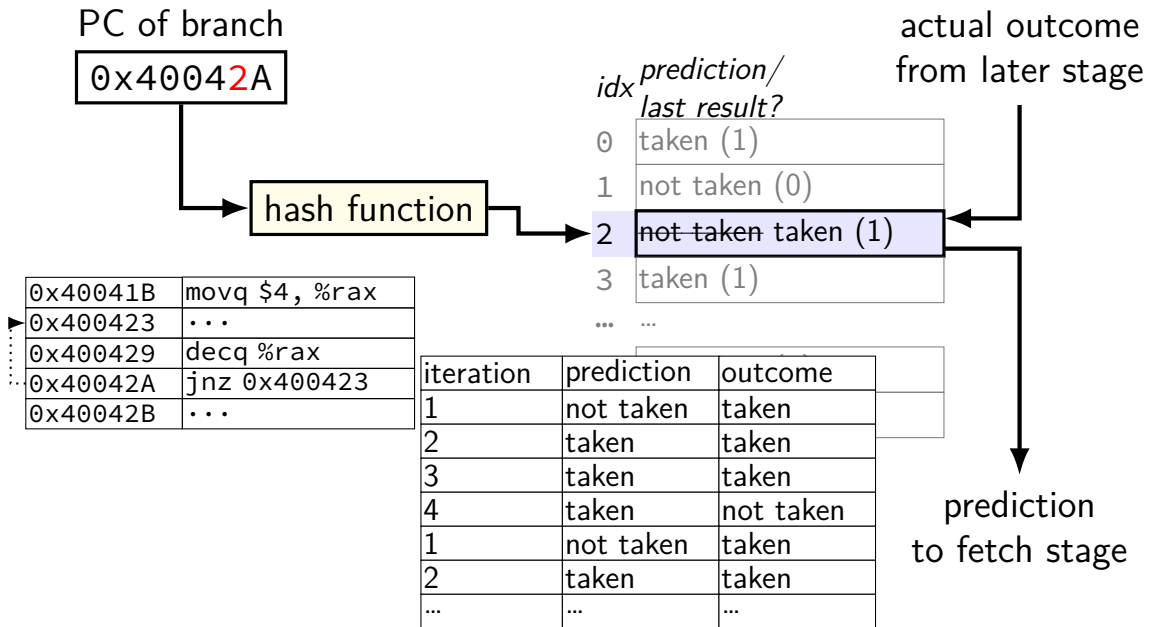
example



example



example



collisions?

two branches could have same hashed PC

nothing in table tells us about this

versus direct-mapped cache: had *tag bits* to tell

is it worth it?

adding tag bits makes table *much* larger and/or slower

but does anything go wrong when there's a collision?

collision results

possibility 1: both branches usually taken

no actual conflict — prediction is better(!)

possibility 2: both branches usually not taken

no actual conflict — prediction is better(!)

possibility 3: one branch taken, one not taken

performance probably worse

1-bit predictor for loops

predicts first and last iteration wrong

example: branch to beginning — but same for branch from beginning to end

everything else correct

exercise (pt 1)

use 1-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate for $i \% 3 == 0$?

```
int i = 0;
while (true) {
    if (i % 3 == 0)
        goto next;
    ...
next:
    i += 1;
    if (i == 50)
        break;
}
```

exercise (pt 1)

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```
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    if (i % 3 == 0)
        goto next;
    ...
next:
    i += 1;
    if (i == 50)
        break;
}
```

i =	branch	pred	outcome	correct?
0	mod 3	???	T	???
1	mod 3	T	F	no
2	mod 3	F	F	yes
3	mod 3	F	T	no
...	...	...	...	...

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1	mod 3	T	F	no
2	mod 3	F	F	yes
3	mod 3	F	T	no
...	...	...	...	...

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    ...
next:
    i += 1;
    if (i == 50)
        break;
}
```

i =	branch	pred	outcome	correct?
0	mod 3	???	T	???
1	mod 3	T	F	no
2	mod 3	F	F	yes
3	mod 3	F	T	no
...	...	...	...	...

exercise (pt 2)

use 1-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate for `i == 50`?

```
int i = 0;
while (true) {
    if (i % 3 == 0)
        goto next;
    ...
next:
    i += 1;
    if (i == 50)
        break;
}
```


exercise (full)

use 1-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate?

```
int i = 0;
while (true) {
    if (i % 3 == 0)
        goto next;
    ...
next:
    i += 1;
    if (i == 50)
        break;
}
```

exercise (full)

use 1-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate?

```
int i = 0;
while (true) {
    if (i % 3 == 0)
        goto next;
    ...
next:
    i += 1;
    if (i == 50)
        break;
}
```

i =	branch	pred	outcome	correct?
0	mod 3	???	T	???
1	== 50	???	F	???
1	mod 3	T	F	—
2	== 50	F	F	✓
...	...	...	...	...

exercise (full)

use 1-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate?

```
int i = 0;
while (true) {
    if (i % 3 == 0)
        goto next;
    ...
next:
    i += 1;
    if (i == 50)
        break;
}
```

i =	branch	pred	outcome	correct?
0	mod 3	???	T	???
1	== 50	???	F	???
1	mod 3	T	F	—
2	== 50	F	F	✓
...	...	...	...	...

branch target buffer

what if we can't decode LABEL from machine code for `jmp LABEL` or `jle LABEL` fast?

will happen in more complex pipelines

what if we can't decode that there's a `RET`, `CALL`, etc. fast?

BTB: cache for branch targets

idx	valid	tag	ofst	type	target	(more info?)
0x00	1	0x400	5	Jxx	0x3FFFF3	...
0x01	1	0x401	C	JMP	0x401035	----
0x02	0	---	---	---	---	----
0x03	1	0x400	9	RET	---	...
...	...	...	...	...	...	...
0xFF	1	0x3FF	8	CALL	0x404033	...

valid	...
1	...
0	...
0	...
0	...
...	...
0	...

```
0x3FFFF3:  movq %rax, %rsi
0x3FFFF7:  pushq %rbx
0x3FFFF8:  call 0x404033
0x400001:  popq %rbx
0x400003:  cmpq %rbx, %rax
0x400005:  jle 0x3FFFF3
...
0x400031:  ret
...
```

BTB: cache for branch targets

idx	valid	tag	ofst	type	target	(more info?)
0x00	1	0x400	5	Jxx	0x3FFFF3	...
0x01	1	0x401	C	JMP	0x401035	----
0x02	0	---	---	---	---	----
0x03	1	0x400	9	RET	---	...
...	...	...	...	...	...	...
0xFF	1	0x3FF	8	CALL	0x404033	...

valid	...
1	...
0	...
0	...
0	...
...	...
0	...

```

0x3FFFF3:  movq %rax, %rsi
0x3FFFF7:  pushq %rbx
0x3FFFF8:  call 0x404033
0x400001:  popq %rbx
0x400003:  cmpq %rbx, %rax
0x400005:  jle 0x3FFFF3
...
0x400031:  ret
...
```

BTB: cache for branch targets

idx	valid	tag	ofst	type	target	(more info?)
0x00	1	0x400	5	Jxx	0x3FFFF3	...
0x01	1	0x401	C	JMP	0x401035	---
0x02	0	---	---	---	---	---
0x03	1	0x400	9	RET	---	...
...	...	...	...	...	...	...
0xFF	1	0x3FF	8	CALL	0x404033	...

valid	...
1	...
0	...
0	...
0	...
...	...
0	...

```
0x3FFFF3:  movq %rax, %rsi
0x3FFFF7:  pushq %rbx
0x3FFFF8:  call 0x404033
0x400001:  popq %rbx
0x400003:  cmpq %rbx, %rax
0x400005:  jle 0x3FFFF3
...
0x400031:  ret
...
```

indirect branch prediction

`jmp *%rax` or `jmp *(%rax, %rcx, 8)`

BTB can provide a prediction

but can do better with more context

example—predict based on other recent computed jumps
good for polymorphic method calls

table lookup with `Hash(last few jmps)`
instead of `Hash(this jmp)`

beyond 1-bit predictor

devote *more space* to storing history

main goal: rare exceptions don't immediately change prediction

example: branch taken 99% of the time

1-bit predictor: wrong about 2% of the time

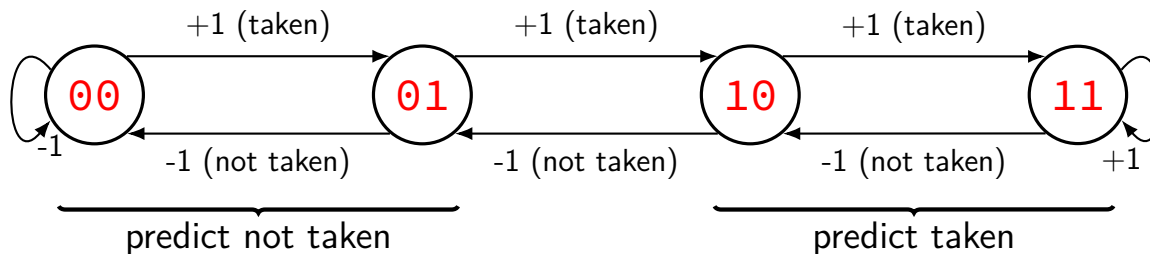
- 1% when branch not taken

- 1% of taken branches right after branch not taken

new predictor: wrong about 1% of the time

- 1% when branch not taken

2-bit saturating counter



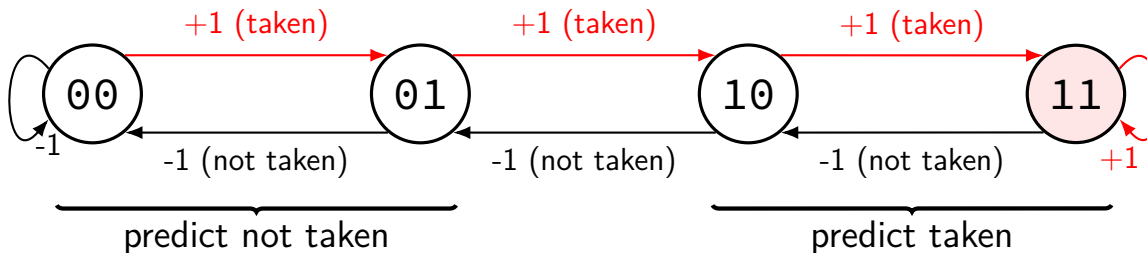
PC of branch

0x40042A

hash function

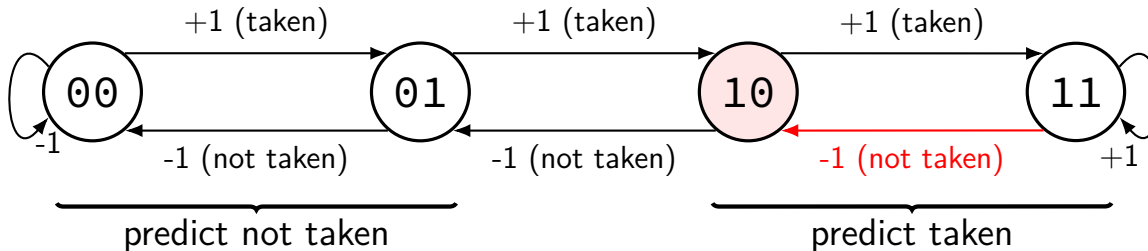
index	counter
0	11
1	01
2	11
...	...
14	10
15	00

2-bit saturating counter



branch always taken:
value increases to 'strongest' taken value

2-bit saturating counter



branch almost always taken, then not taken once:
still predicted as taken

example

0x40041B	movq \$4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

iter.	table before	prediction	outcome	table after
1	01	not taken	taken	10
2	10	taken	taken	11
3	11	taken	taken	11
4	11	taken	not taken	10
1	10	taken	taken	11
2	11	taken	taken	11
3	11	taken	taken	11
4	11	taken	not taken	10
1	10	taken	taken	11
...	...	...	...	...

generalizing saturating counters

2-bit counter: ignore one exception to taken/not taken

3-bit counter: ignore more exceptions

000 $\leftrightarrow$ 001 $\leftrightarrow$ 010 $\leftrightarrow$ 011 $\leftrightarrow$ 100 $\leftrightarrow$ 101 $\leftrightarrow$ 110 $\leftrightarrow$ 111

000-011: not taken

100-111: taken

check_passphrase

```
int check_passphrase(const char *versus) {  
    int i = 0;  
    while (passphrase[i] == versus[i] &&  
           passphrase[i]) {  
        i += 1;  
    }  
    return (passphrase[i] == versus[i]);  
}
```

number of iterations = number matching characters

leaks information about passphrase, oops!

exploiting check_passphrase (1)

guess	measured time
aaaa	100 ± 5
baaa	103 ± 4
caaa	102 ± 6
daaa	111 ± 5
aaaa	99 ± 6
faaa	101 ± 7
gaaa	104 ± 4
...	...

exploiting check_passphrase (2)

guess	measured time
daaa	102 ± 5
dbaa	99 ± 4
dcaa	104 ± 4
ddaa	100 ± 6
deaa	102 ± 4
dfaa	109 ± 7
dgaa	103 ± 4
...	...

timing and cryptography

lots of asymmetric cryptography uses big-integer math

example: multiplying 500+ bit numbers together

how do you implement that?

big integer multiplication

say we have two 64-bit integers x, y

and want to 128-bit product, but our multiply instruction only does 64-bit products

one way to multiply:

divide x, y into 32-bit parts: $x = x_1 \cdot 2^{32} + x_0$ and $y = y_1 \cdot 2^{32} + y_0$

then $xy = x_1y_12^{64} + x_1y_0 \cdot 2^{32} + x_0y_1 \cdot 2^{32} + x_0y_0$

big integer multiplication

say we have two 64-bit integers x, y

and want to 128-bit product, but our multiply instruction only does 64-bit products

one way to multiply:

divide x, y into 32-bit parts: $x = x_1 \cdot 2^{32} + x_0$ and $y = y_1 \cdot 2^{32} + y_0$

then $xy = x_1y_12^{64} + x_1y_0 \cdot 2^{32} + x_0y_1 \cdot 2^{32} + x_0y_0$

can extend this idea to arbitrarily large numbers

number of smaller multiplies depends on size of numbers!

big integers and cryptography

naive multiplication idea:

- number of steps depends on size of numbers

problem: sometimes the value of the number is a secret

- e.g. part of the private key

oops! revealed through timing

big integer timing attacks in practice (1)

early versions of OpenSSL (TLS implementation) had timing attack

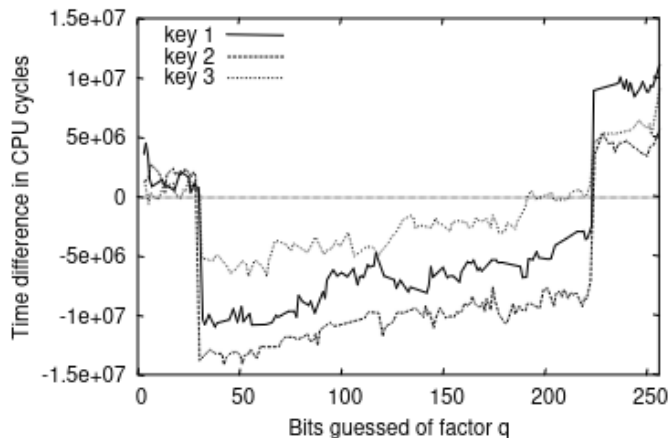
Brumley and Boneh, "Remote Timing Attacks are Practical" (Usenix Security '03)

attacker could figure out bits of private key from timing

why? variable-time multiplication and modulus operations

got faster/slower depending on how input was related to private key

big integer timing attacks in practice (2)



(a) The zero-one gap $T_g - T_{g_{hi}}$ indicates that we can distinguish between bits that are 0 and 1 of the RSA factor q for 3 different randomly-generated keys. For clarity, bits of q that are 1 are omitted, as the x -axis can be used for reference for this case.

browsers and website leakage

web browsers run code from untrusted webpages

one goal: can't tell what other webpages you visit

some webpage leakage (1)

...as you can see [here](#), [here](#), and [here](#) ...

convenient feature 1: browser marks visited links

```
<script>
var the_color = window.getComputedStyle(
    document.querySelector('a[href=~"foo.com"]')
).color
if (the_color == ...) { ... }
</script>
```

convenient feature 2: scripts can query current color of something

some webpage leakage (1)

...as you can see [here](#), [here](#), and [here](#) ...

convenient feature 1: browser marks visited links

```
<script>
var the_color = window.getComputedStyle(
    document.querySelector('a[href=~"foo.com"]')
).color
if (the_color == ...) { ... }
</script>
```

~~convenient feature 2: scripts can query current color of something~~

fix 1: getComputedStyle lies about the color

fix 2: limited styling options for visited links

some webpage leakage (2)

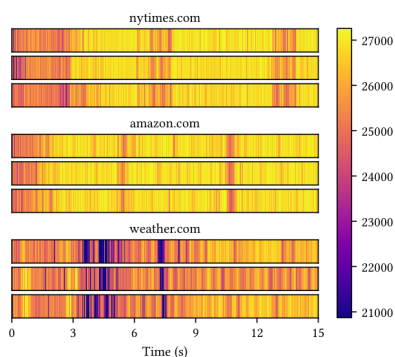
one idea: script in webpage times loop that writes big array

variation in timing depends on other things running on machine

some webpage leakage (2)

one idea: script in webpage times loop that writes big array

variation in timing depends on **other things running on machine**



turns out, other webpages
create distinct “signatures”

Figure from Cook et al, “There’s Always a Bigger Fish: Clarifying Analysis of Machine-Learning-Assisted Side-Channel Attack” (ISCA ’22)

Figure 3: Example loop-counting traces collected over 15 seconds. Darker shades indicate smaller counter values and lower instruction throughput.

inferring cache accesses (1)

suppose I time accesses to array of chars:

reading array[0]: 3 cycles

reading array[64]: 4 cycles

reading array[128]: 4 cycles

reading array[192]: 20 cycles

reading array[256]: 4 cycles

reading array[288]: 4 cycles

...

what could cause this difference?

array[192] not in some cache, but others were

inferring cache accesses (2)

some psuedocode:

```
char array[CACHE_SIZE];  
AccessAllOf(array);  
*other_address += 1;  
TimeAccessingArray();
```

suppose during these accesses I discover that `array[128]` is slower to access

probably because `*other_address` loaded into cache + evicted it

what do we know about `other_address`? (select all that apply)

- A. same cache tag B. same cache index C. same cache offset
- D. diff. cache tag E. diff. cache index F. diff. cache offset

some complications

caches often use physical, not virtual addresses

- (and need to know about physical address to compare index bits)

- (but can infer physical addresses with measurements/asking OS)

- (often OS allocates contiguous physical addresses esp. w/‘large pages’)

storing/processing timings evicts things in the cache

- (but can compare timing with/without access of interest to check for this)

processor “pre-fetching” may load things into cache before access is timed

- (but can arrange accesses to avoid triggering prefetcher and make sure to measure with memory barriers)

some L3 caches use a simple hash function to select index instead of index bits

exercise: inferring cache accesses (1)

```
char *array;  
array = AllocateAlignedPhysicalMemory(CACHE_SIZE);  
LoadIntoCache(array, CACHE_SIZE);  
if (mystery) {  
    *pointer += 1;  
}  
if (TimeAccessTo(&array[index]) > THRESHOLD) {  
    /* pointer accessed */  
}
```

suppose pointer is 0x1000188

and cache (of interest) is direct-mapped, 32768 (2^{15}) byte, 64-byte blocks

what array index should we check?

aside

```
array = AllocateAlignedPhysicalMemory(CACHE_SIZE);  
LoadIntoCache(array, CACHE_SIZE);  
if (mystery) { *pointer += 1; }  
if (TimeAccessTo(&array[index]) > THRESHOLD) {  
    /* pointer accessed */  
}
```

will this detect when pointer accessed? yes

will this detect if mystery is true? not quite

...because branch prediction could started cache access

exercise: inferring cache accesses (2)

```
char *other_array = ...;
char *array;
array = AllocateAlignedPhysicalMemory(CACHE_SIZE);
LoadIntoCache(array, CACHE_SIZE);
other_array[mystery] += 1;
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (TimeAccessTo(&array[i]) > THRESHOLD) {
        /* found something interesting */
    }
}
```

other_array at 0x200400, and interesting index is $i=0x800$, then what was mystery?

exercise: inferring cache accesses (2)

```
char *array;  
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);  
LoadIntoCache(array, CACHE_SIZE);  
if (mystery) {  
    *pointer = 1;  
}  
if (TimeAccessTo(&array[index1]) > THRESHOLD ||  
    TimeAccessTo(&array[index2]) > THRESHOLD) {  
    /* pointer accessed */  
}
```

pointer is 0x1000188

cache is 2-way, 32768 (2^{15}) byte, 64-byte blocks, ??? replacement

what array indexes should we check?

PRIME+PROBE

name in literature: PRIME + PROBE

PRIME: fill cache (or part of it) with values

do thing that uses cache

PROBE: access those values again and see if it's slow

(one of several ways to measure how cache is used)

coined in attacks on AES encryption

example: AES (1)

from Osvik, Shamir, and Tromer, “Cache Attacks and Countermeasures: the Case of AES” (2004)

early AES implementation used lookup tables

goal: detect index into lookup table

index depended on key + data being encrypted

tricks they did to make this work

vary data being encrypted

subtract average time to look for what changes

lots of measurements

example: AES (2)

from Osvik, Shamir, and Tromer, “Cache Attacks and Countermeasures: the Case of AES” (2004)

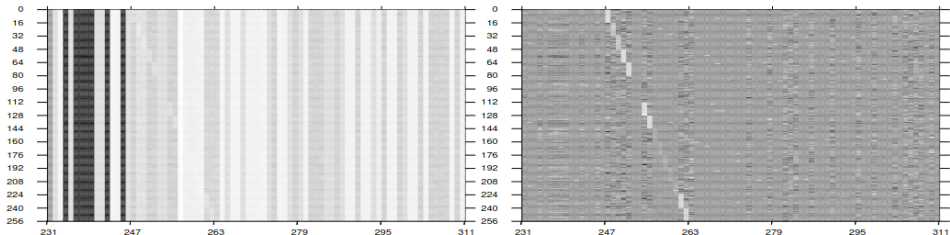


Fig. 5. Prime+Probe attack using 30,000 encryption calls on a 2GHz Athlon 64, attacking Linux 2.6.11 `dm-crypt`. The horizontal axis is the evicted cache set (i.e., $\langle y \rangle$ plus an offset due to the table's location) and the vertical axis is p_0 . Left: raw timings (lighter is slower). Right: after subtraction of the average timing of the cache set. The bright diagonal reveals the high nibble of $p_0 = 0x00$.

reading a value

```
char *array;  
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);  
AccessAllOf(array);  
other_array[mystery * BLOCK_SIZE] += 1;  
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {  
    if (CheckIfSlowToAccess(&array[i])) {  
        ...  
    }  
}
```

with 32KB direct-mapped cache

suppose we find out that `array[0x400]` is slow to access

and `other_array` starts at address `0x100000`

what was `mystery`?

revisiting an earlier example (1)

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
LoadIntoCache(array, CACHE_SIZE);
if (mystery) {
    *pointer += 1;
}
if (TimeAccessTo(&array[index]) > THRESHOLD) {
    /* pointer accessed */
}
```

what if mystery is false *but* branch mispredicted?

revisiting an earlier example (2)

	cycle #	0	1	2	3	4	5	6	7	8	9	10	11
movq mystery, %rax		F	D	R	I	E	E	E	W	C			
test %rax, %rax		F	D	R				I	E	W	C		
jz skip (mispred.)		F	D	R				I	E	W	C		
mov pointer, %rax		F	D	R	I	E	E	E	W				
mov (%rax), %r8			F	D	R				I	E	W		
add \$1, %r8			F	D	R								
mov %r8, %rax				F	D	R							
...													
skip: ...									F	D	R		

avoiding/triggering this problem

```
if (something false) {  
    access *pointer;  
}
```

what can we do to make access more/less likely to happen?

reading a value without really reading it

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
if (something false) {
    other_array[mystery * BLOCK_SIZE] += 1;
}
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

if branch mispredicted, cache access may **still happen**

can find the value of `mystery`

seeing past a segfault? (1)

```
Prime();  
if (something false) {  
    triggerSegfault();  
    Use(*pointer);  
}  
Probe();
```

could cache access for `*pointer` still happen?

yes, if:

- branch for if statement mispredicted, and
- `*pointer` starts before segfault detected

seeing past a segfault? (2)

operations in virtual memory lookup:

- translate virtual to physical address

- check if access is permitted by permission bits

Intel processors: looks like these were separate steps, so...

```
Prime();
```

```
if (something false) {
```

```
    int value = ReadMemoryMarkedNonReadableInPageTable();
```

```
    access other_array[value * ...];
```

```
}
```

```
Probe();
```

seeing past a segfault? (2)

operations in virtual memory lookup:

- translate virtual to physical address

- check if access is permitted by permission bits

Intel processors: looks like these were separate steps, so...

```
Prime();
```

```
if (something false) {
```

```
    int value = ReadMemoryMarkedNonReadableInPageTable();
```

```
    access other_array[value * ...];
```

```
}
```

```
Probe();
```

seeing past a segfault? (2)

operations in virtual memory lookup:

- translate virtual to physical address

- check if access is permitted by permission bits

Intel processors: looks like these were separate steps, so...

```
Prime();
```

```
if (something false) {
```

```
    int value = ReadMemoryMarkedNonReadableInPageTable();
```

```
    access other_array[value * ...];
```

```
}
```

```
Probe();
```

seeing past a segfault? (2)

operations in virtual memory lookup:

- translate virtual to physical address

- check if access is permitted by permission bits

Intel processors: looks like these were separate steps, so...

```
Prime();
```

```
if (something false) {
```

```
    int value = ReadMemoryMarkedNonReadableInPageTable();
```

```
    access other_array[value * ...];
```

```
}
```

```
Probe();
```


Meltdown

from Lipp et al, "Meltdown: Reading Kernel Memory from User Space"

```
// %rcx = kernel address  
// %rbx = array to load from to cause eviction  
xor %rax, %rax      // rax ← 0
```

retry:

```
// rax ← memory[kernel address] (segfaults)  
// but check for segfault done out-of-order on Intel  
movb (%rcx), %al  
// rax ← memory[kernel address] * 4096 [speculated]  
shl $0xC, %rax  
jz retry             // not-taken branch  
// access array[memory[kernel address] * 4096]  
mov (%rbx, %rax), %rbx
```

Meltdown

from Lipp et al, "Meltdown: Reading Kernel Memory from User Space"

```
// %rcx = kernel address
// %rbx = array address
xor %rax, %rax
retry:
    // rax <- memory[kernel address] * 4096 [speculated]
    // but check for segfault done out-of-order on Intel
    movb (%rcx), %al
    // rax <- memory[kernel address] * 4096 [speculated]
    shl $0xC, %rax
    jz retry
    // access array[memory[kernel address] * 4096]
    mov (%rbx, %rax), %rbx
```

space out accesses by 4096
ensure separate cache sets and
avoid triggering prefetcher

Meltdown

from Lipp et al, "Meltdown: Reading Kernel Memory from User Space"

```
// %rcx = kernel address
// %rbx = 0
xor %rax, %rax
retry:
    // rax <- memory[kernel address] * 4096 [speculated]
    // but check for segfault done out-of-order on Intel
    movb (%rcx), %al
    // rax <- memory[kernel address] * 4096 [speculated]
    shl $0xC, %rax
    jz retry // not-taken branch
    // access array[memory[kernel address] * 4096]
    mov (%rbx, %rax), %rbx
```

repeat access if zero
apparently value of zero speculatively read
when real value not yet available

Meltdown

from Lipp et al, "Meltdown: Reading Kernel Memory from User Space"

```
// %rcx = kernel address
// %rbx = ...
xor %rax, %rax
retry:
// rax <- ...
// but check for segfault done out-of-order on Intel
movb (%rcx), %al
// rax <- memory[kernel address] * 4096 [speculated]
shl $0xC, %rax
jz retry // not-taken branch
// access array[memory[kernel address] * 4096]
mov (%rbx, %rax), %rbx
```

Meltdown

from Lipp et al, "Meltdown: Reading Kernel Memory from User Space"

```
// %rcx = kernel address
```

segfault actually happens eventually

option 1: okay, just start a new process every time

option 2: way of suppressing exception (transactional memory support)

```
// but check for segfault done out-of-order on Intel
```

```
movb (%rcx), %al
```

```
// rax <- memory[kernel address] * 4096 [speculated]
```

```
shl $0xC, %rax
```

```
jz retry // not-taken branch
```

```
// access array[memory[kernel address] * 4096]
```

```
mov (%rbx, %rax), %rbx
```

Meltdown fix

HW: permissions check done with/before physical address lookup
was already done by AMD, ARM apparently?
now done by Intel

SW: separate page tables for kernel and user space
don't have sensitive kernel memory pointed to by page table
when user-mode code running
unfortunate performance problem
exceptions start with code that switches page tables

reading a value without really reading it

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
if (something false) {
    other_array[mystery * BLOCK_SIZE] += 1;
}
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

if branch mispredicted, cache access may **still happen**

can find the value of `mystery`

mistraining branch predictor?

```
if (something) {  
    CodeToRunSpeculatively()  
}
```

how can we have 'something' be false, but predicted as true

run lots of times with something true

then do actually run with something false

contrived(?) vulnerable code (1)

suppose this C code is run with extra privileges

(e.g. in system call handler, library called from JavaScript in webpage, etc.)

assume x chosen by attacker

(example from original Spectre paper)

```
if (x < array1_size)
    y = array2[array1[x] * 4096];
```

the out-of-bounds access (1)

```
char array1[...];
```

```
...
```

```
int secret;
```

```
...
```

```
y = array2[array1[x] * 4096];
```

suppose array1 is at 0x10000000 and

secret is at 0x103F0003;

what x do we choose to make array1[x] access first byte of secret?

the out-of-bounds access (2)

```
unsigned char array1[...];
```

```
...
```

```
int secret;
```

```
...
```

```
y = array2[array1[x] * 4096];
```

suppose our cache has 64-byte blocks and 8192 sets

and `array2[0]` is stored in cache set 0

if the above evicts something in cache set 128,
then what do we know about `array1[x]`?

the out-of-bounds access (2)

```
unsigned char array1[...];
```

```
...
```

```
int secret;
```

```
...
```

```
y = array2[array1[x] * 4096];
```

suppose our cache has 64-byte blocks and 8192 sets

and `array2[0]` is stored in cache set 0

if the above evicts something in cache set 128,
then what do we know about `array1[x]`?

is 2 or 130

exploit with contrived(?) code

```
/* in kernel: */  
int systemCallHandler(int x) {  
    if (x < array1_size)  
        y = array2[array1[x] * 4096];  
    return y;  
}
```

```
/* exploiting code */  
    /* step 1: mistrain branch predictor */  
for (a lot) {  
    systemCallHandler(0 /* less than array1_size */);  
}  
    /* step 2: evict from cache using misprediction */  
Prime();  
systemCallHandler(targetAddress - array1Address);  
int evictedSet = ProbeAndFindEviction();  
int targetValue = (evictedSet - array2StartSet) / setsPer4K;
```

really contrived?

```
char *array1; char *array2;  
if (x < array1_size)  
    y = array2[array1[x] * 4096];
```

times 4096 shifts so we can get lower bits of target value
so all bits effect what cache block is used

really contrived?

```
char *array1; char *array2;  
if (x < array1_size)  
    y = array2[array1[x] * 4096];
```

times 4096 shifts so we can get lower bits of target value
so all bits effect what cache block is used

```
int *array1; int *array2;  
if (x < array1_size)  
    y = array2[array1[x]];
```

will still get *upper* bits of array1[x] (can tell from cache set)

can still read arbitrary memory!

want memory at 0x10000?

upper bits of 4-byte integer at 0x0FFFE

bounds check in kernel

```
if (x < array1_size) {  
    y = array2[array1[x]];  
}
```

our template

```
void SomeSystemCallHandler(int index) {  
    if (index > some_table_size)  
        return ERROR;  
    int kind = table[index];  
    switch (other_table[kind].foo) {  
        ...  
    }  
}
```

actual code

bounds check in kernel

```
if (x < array1_size) {  
    y = array2[array1[x]];  
}
```

our template

```
void SomeSystemCallHandler(int index) {  
    if (index > some_table_size)  
        return ERROR;  
    int kind = table[index];  
    switch (other_table[kind].foo) {  
        ...  
    }  
}
```

actual code

bounds check in kernel

```
if (x < array1_size) {  
    y = array2[array1[x]];  
}
```

our template

```
void SomeSystemCallHandler(int index) {  
    if (index > some_table_size)  
        return ERROR;  
    int kind = table[index];  
    switch (other_table[kind].foo) {  
        ...  
    }  
}
```

actual code

bounds check in kernel

```
if (x < array1_size) {  
    y = array2[array1[x]];  
}
```

our template

```
void SomeSystemCallHandler(int index) {  
    if (index > some_table_size)  
        return ERROR;  
    int kind = table[index];  
    switch (other_table[kind].foo) {  
        ...  
    }  
}
```

actual code

exercise

```
char *array;
// PRIME
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
// (some code we don't control)
other_array[mystery * BLOCK_SIZE] += 1;
// PROBE
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

64KB (2^{16} B) direct-mapped cache with 64B blocks

array[0x800] slow to access;

other_array at 0x40000000

value of mystery?

exercise solution (1)

$\text{NUM_SETS} = 64\text{KB}/64\text{B} = 1\text{K} (1024) \text{ sets}$

$\text{array}[0\text{x}800]$ has cache index $0\text{x}800/\text{BLOCK_SIZE} \bmod \text{NUM_SETS}$
 $= \text{cache index } 32$

know $\text{other_array}[\text{mystery} * \text{BLOCK_SIZE}]$ had same index

$\text{other_array}[0]$ at cache index 0
 $(0\text{x}4000000 / \text{BLOCK_SIZE}) \bmod \text{NUM_SETS} = 0$

exercise solution (2)

recall have found:

other_array[0] at index 0;

other_array[mystery*BLOCK_SIZE] has index 32 (same as array[0x800])

other_array[X] at cache index $(0 + X/\text{BLOCK_SIZE} \bmod \text{NUM_SETS})$

advanced by $X/\text{BLOCK_SIZE}$ blocks

wrapping around after NUM_SETS blocks

$$X = \text{mystery} * \text{BLOCK_SIZE}$$

$$32 = 0 + \text{mystery} \bmod \text{NUM_SETS}$$

$$\text{mystery} = 32 \text{ or } 32 \pm 1024 \text{ or } 32 \pm 1024 \times 2 \text{ or etc.}$$

exercise

```
char *array;
//PRIME
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
other_array[mystery] += 1;
//PROBE
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

with 64KB direct-mapped cache with 64B blocks

suppose we find out that `array[0x200]` is slow to access

and `other_array` starts at some multiple of cache size

What was mystery?

```

char *array;
//PRIME
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array); // PRIME
other_array[mystery] += 1;
//PROBE
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) // PROBE
        {...}
}

```

- $NSETS = CACHE_SIZE / BLOCK_SIZE = 64KB / 64B = 1K = 2^{10}$
- And this affected `array[0x200]`
 - Which had cache index $0x200 / BLOCK_SIZE = 512 / 64 = 8$
 - Or `0b 0010 0000 0000`
- `other_array[mystery] = other_array + mystery` (because these are char array)
- If we know the base address of `other_array` is `0x20000`, we need to index $(0x20000 + mystery) = 8$
- `0b 0010 0000 0000 0000 0000 //other_array`
- `+0b 0000 0000 0000 0000 //mystery`
- `=0b 0010 0000 0010 0000`
- So we get a couple bits in the low-order byte of `mystery` and the next byte

not just BLOCK_SIZE

```
char *array, *other_array;
// PRIME
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
// (some code we don't control)
other_array[mystery * N] += 1; // previously: * BLOCK_SIZE
// PROBE
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

64KB (2^{16} B) direct-mapped cache with 64B blocks

array[0x800] slow to access?

other_array at 0x40000000 (index 0, offset 0)

value of mystery if $N = 1$? $N = 32 * 64$?

solution (N=1)

$$\lfloor \text{mystery} * N / \text{BLOCK_SIZE} \rfloor \bmod 1024 = 32$$

$$\lfloor \text{mystery} * N / \text{BLOCK_SIZE} \rfloor = 32 + 1024K$$

let offset be some number in $[0, \text{BLOCK_SIZE})$:

$$\text{mystery} * N = \text{BLOCK_SIZE} \times (32 + 1024Z) + \text{offset}$$

$$\text{mystery} = \text{BLOCK_SIZE} \times (32 + 1024Z) + N \times \text{offset}$$

$$\text{mystery} = 64 \times (32 + 1024Z) + N \times \text{offset}$$

N=1: mystery = 2048, 2049, 2050, ..., 2048 + 63, $64 \cdot 1024 + 2048$,
 $64 \cdot 1024 + 2048 + 1$, ...

exercise (N=32*64)

what if $N = 32 \cdot 64$

recall: `other_array[0]` is set 0, offset 0

`other_array[mystery * N]` is set 32

possible values of mystery?

$$\begin{aligned}\text{mystery} \cdot 32 \cdot 64 &= 64(32 + 1024Z) + \text{offset} \\ &= 64 \cdot 32 + 65536Z + \text{offset}\end{aligned}$$

$$\text{mystery} = 1 + \frac{65536}{64 \cdot 32}Z + \frac{\text{offset}}{64 \cdot 32} = 1 + 32Z$$

alternate view

learn index bits of mystery * N

this example: bits 6–15

$N = 1$, bits 6–15 of mystery

$N = 64$, bits 0–9 of mystery

$N = 32 * 64$ (2^{11}), bits 0–4 of mystery

variation: different starting location

other_array starts at 0x4001440

then other_array[0] at cache index

$$0x4001440 / \text{BLOCK_SIZE} \bmod \text{NUM_SETS} = 51$$

$$(51 + \text{mystery} * \text{BLOCK_SIZE} / \text{BLOCK_SIZE}) \bmod \text{NUM_SETS} = 32$$

mystery = -19 or 1005 or 2029 or ...

variation: associative cache

```
char *array;  
// PRIME  
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);  
AccessAllOf(array);  
// (some code we don't control)  
other_array[mystery * BLOCK_SIZE] += 1;  
// PROBE  
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {  
    if (CheckIfSlowToAccess(&array[i])) { ... }  
}
```

suppose 2-way 64KB cache instead of direct-mapped

$\text{NUM_SETS} = 64\text{KB} / 2 / 64\text{B} = 512$ sets

array[0x800] still has cache index 32 (still)

but now mystery can be 32 or $32 + 512$ or $32 + 512 \cdot 2$ or ...

variation: associative cache (2)

```
char *array;  
// PRIME  
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);  
AccessAllOf(array);  
// (some code we don't control)  
other_array[mystery * BLOCK_SIZE] += 1;  
// PROBE  
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {  
    if (CheckIfSlowToAccess(&array[i])) { ... }  
}
```

suppose 2-way 64KB cache w/ 64B and `array[0x8800]` is slow

$$0x8800 / \text{BLOCK_SIZE} = 544 = 512 + 32$$

since 512 sets total, still set index 32

mystery still 32 or $32 + 512$ or $32 + 512 \cdot 2$ or ...

exercise

if 4-way 64KB cache w/64B blocks and something from cache set 32 evicted,
then where could slow access be?

recall: 2-way cache: $i=0x800$, $i=0x8800$

- A. $i=0x400$, $i=0x800$, $i=0x8400$, $i=0x8800$
- B. $i=0x800$, $i=0x8800$, $i=0x10800$, $i=0x18800$
- C. $i=0x800$, $i=0x4800$, $i=0x8800$, $i=0xc800$
- D. $i=0x800$, $i=0x4800$, $i=0x8800$, $i=0x10800$
- E. something else

EVICT+RELOAD

PRIME+PROBE: fill cache, detect eviction

alternate idea EVICT+RELOAD:

```
unsigned char *probe_array;  
posix_memalign(&probe_array, CACHE_SIZE, CACHE_SIZE);  
access OTHER things to evict all of probe_array  
if (something false) {  
    read probe_array[mystery * BLOCK_SIZE];  
}
```

check which value from probe_array is faster

requires code to access something you can access

but often easier to setup/more reliable than PRIME+PROBE

EVICT+RELOAD

PRIME+PROBE: fill cache, detect eviction

alternate idea **EVICT**+RELOAD:

```
unsigned char *probe_array;  
posix_memalign(&probe_array, CACHE_SIZE, CACHE_SIZE);  
access OTHER things to evict all of probe_array  
if (something false) {  
    read probe_array[mystery * BLOCK_SIZE];  
}
```

check which value from probe_array is faster

requires code to access something you can access

but often easier to setup/more reliable than PRIME+PROBE

EVICT+RELOAD

PRIME+PROBE: fill cache, detect eviction

alternate idea EVICT+RELOAD:

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if (something false) {  
    read probe_array[mystery * BLOCK_SIZE];  
}
```

check which value from probe_array is faster

requires code to access something you can access

but often easier to setup/more reliable than PRIME+PROBE

into exploit: Meltdown

```
uint8_t* probe_array = new uint8_t[256 * 4096];  
// ... Make sure probe_array is not cached  
uint8_t kernel_memory_val = *(uint8_t*)(kernel_address);  
uint64_t final_kernel_memory = kernel_memory_val * 4096;  
uint8_t dummy = probe_array[final_kernel_memory];  
// ... catch page fault  
// ... in signal handler, determine which of 256 slots in probe_array
```

privilege levels?

vulnerable code runs with higher privileges

so far: higher privileges = kernel mode

but other common cases of higher privileges

example: scripts in web browsers

JavaScript

JavaScript: scripts in webpages

not supposed to be able to read arbitrary memory, but...

can access arrays to examine caches

and could take advantage of some browser function being vulnerable

JavaScript

JavaScript: scripts in webpages

not supposed to be able to read arbitrary memory, but...

can access arrays to examine caches

and could take advantage of some browser function being vulnerable

or — doesn't even need browser to supply vulnerable code itself!

just-in-time compilation?

for performance, compiled to machine code, run in browser

not supposed to be access arbitrary browser memory

example JavaScript code from paper:

```
if (index < simpleByteArray.length) {  
    index = simpleByteArray[index | 0];  
    index = (((index * 4096) | 0) & (32*1024*1024-1)) | 0;  
    localJunk ^= probeTable[index|0]|0;  
}
```

web page runs a lot to train branch predictor

then does run with out-of-bounds index

examines what's evicted by probeTable access

supplying own attack code?

JavaScript: could supply own attack code

turns out also possible with kernel mode scenario

trick: don't need to *actually run* code

...just need branch predictor to fetch it!

other misprediction

so far: talking about mispredicting direction of branch

what about mispredicting target of branch in, e.g.:

```
// possibly from C code like:  
//    (*function_pointer)();  
jmp *%rax
```

```
// possibly from C code like:  
//    switch(rcx) { ... }  
jmp *(%rax,%rcx,8)
```

an idea for predicting indirect jumps

for jumps like `jmp *%rax` predict target with cache:

bottom 12 bits of jmp address	last seen target
0x0–0x7	0x200000
0x8–0xF	0x440004
0x10–0x18	0x4CD894
0x18–0x20	0x510194
0x20–0x28	0x4FF194
...	...
0xFF8–0xFFFF	0x3F8403

Intel Haswell CPU did something similar to this

uses bits of last several jumps, not just last one

can mistrain this branch predictor

using mispredicted jump

- 1: find some kernel function with `jmp *%rax`
- 2: mistrain branch target predictor for it to jump to chosen code
use code at address that conflicts in “recent jumps cache”
- 3: have chosen code be attack code (e.g. array access)
either write special code OR
find suitable instructions (e.g. array access) in existing kernel code

Spectre variants

showed Spectre variant 1 (array bounds), 2 (indirect jump)
from original paper

other possible variations:

- could cause other things to be mispredicted

 - prediction of where functions return to?

 - values instead of which code is executed?

- could use side-channel other than data cache changes

 - instruction cache

 - cache of pending stores not yet committed

 - contention for resources on multi-threaded CPU core

 - branch prediction changes

 - ...

some Linux kernel mitigations (1)

replace `array[x]` with
`array[x & ComputeMask(x, size)]`

...where `ComputeMask()` returns

0 if $x > \text{size}$

0xFFFF...F if $x \leq \text{size}$

...and `ComputeMask()` does not use jumps:

```
mov x, %r8
mov size, %r9
cmp %r9, %r8
sbb %rax, %rax // sbb = subtract with borrow
                // either 0 or -1
```

some Linux kernel mitigations (2)

for indirect branches:

with hardware help:

- separate indirect (computed) branch prediction for kernel v user mode
- other branch predictor changes to isolate better

without hardware help:

- transform `jmp *(%rax)`, etc. into code that will only be predicted to jump to safe locations (by writing assembly very carefully)

only safe prediction

as replacement for `jmp *(%rax)`

code from Intel's "Retpoline: A Branch Target Injection Mitigation"

```
    call load_label
capture_ret_spec:    /* <-- want prediction to go here */
    pause
    lfence
    jmp capture_ret_spec
load_label:
    mov %rax, (%rsp)
    ret
```


backup slides

Quiz week 14

Question 5 (0 / 4 pt; mean 0.16)

Consider the following C code:

```
struct Files all_files[NUM_FILES];
int GetLastByteOfFile(int index) {
    if (index >= 0 && index < all_files) {
        struct File *file = &all_files[index];
        if (file->type == MEMORY) {
            return file->data[file->size - 1];
        } else if (file->type == DISK) {
            return GetLastByteOfDiskFile(file);
        } else {
            return -1;
        }
    } else {
        return -1;
    }
}
```

If the above function runs in kernel mode, we might be able to use a Spectre-style attack where the cache evictions caused by memory access of `file->data[file.size - 1]` allows us learn about the value of an arbitrary memory location. To perform this attack, the attacker would prefer to choose an out-of-bounds index such that ____.

- A. 88% ☐ the address of `all_files[index].data[file.size - 1]` is the memory address whose value they want to learn about
- B. 2% ☐ the address of `all_files[index].type` is the memory address they want to learn about
- C. 4% ^T
(correct) ☒ the address of `all_files[index].size` is the memory address they want to learn about
- D. ☐ the value of `all_files[index].type` is DISK

- `file = &all_files[index]`
- So if we provide an out of bounds index, we can read arbitrary memory
- `file->data[]`, ie `all_files[index].data[]` is like array2
- `file->size`, ie `all_files[index].size`, is like array1

```
if (x < array1_size) {  
    y = array2[array1[x]];  
}
```

our template

```
void SomeSystemCallHandler(int index) {  
    if (index > some_table_size)  
        return ERROR;  
    int kind = table[index];  
    switch (other_table[kind].foo) {  
        ...  
    }  
}
```

actual code

- kind ~ file->size, ie all_files[index].size, is like array1
 - This is the address we want to learn about by observing its cache behavior
- other_table [] ~ file->data[], ie all_files[index].data[] is like array2
 - Not using the .foo here

Question 5 (0 / 4 pt; mean 0.16)

Consider the following C code:

```
struct Files all_files[NUM_FILES];
int GetLastByteOfFile(int index) {
    if (index >= 0 && index < all_files) {
        struct File *file = &all_files[index];
        if (file->type == MEMORY) {
            return file->data[file->size - 1];
        } else if (file->type == DISK) {
            return GetLastByteOfDiskFile(file);
        } else {
            return -1;
        }
    } else {
        return -1;
    }
}
```

If the above function runs in kernel mode, we might be able to use a Spectre-style attack where the cache evictions caused by memory access of `file->data[file.size - 1]` allows us learn about the value of an arbitrary memory location. To perform this attack, the attacker would prefer to choose an out-of-bounds index such that ____.

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- C. 4% ^T
(correct) ☒ the address of `all_files[index].size` is the memory address they want to learn about
- D. ☐ the value of `all_files[index].type` is DISK

• Why not A?

• `all_files[index].data` is like `array2`

- Based on which cache set is affected by different index values, we learn what those index values are
- So we need to set up the index (~array1) to refer to the memory location we're interested in – here `file.size`, ie `all_files[index].size`

Q4

Consider the following code:

```
unsigned char check_array[32768];
int mystery = /* unknown */;

int Check(int key) {
    return check_array[(key + mystery) % 32768];
}
```

Suppose that:

- (to simplify the problem) virtual memory is not use
- check_array is located at physical address 0x1400000
- the system has a 2-way 64KB (2 to 16 byte) data cache with 64-byte cache blocks.
- Check is compiled to perform exactly three memory accesses:
 - to read the global variable read mystery
 - to reads its return address from the stack
 - to read from check_array

There was a typo originally that we fixed.
Originally wrote %16384 instead of %32768,
and 4096 instead of 8240 for the last row

Suppose we determine that calling Check evicts from the data cache sets as follows:

key value | evicts values from cache set indexes

0	15, 72, 435
32	15, 72, 435
48	15, 72, 436
128	15, 72, 437
8240	15, 52, 72

Based on this information what is a possible value for mystery?

- $(0 + \text{mystery}) // 64 \text{ +/- multiple of } 512 \text{ (number of cache sets)} = 435$
- $(32 + \text{mystery}) // 64 \text{ +/- multiple of } 512 \text{ (number of cache sets)} = 435$
- $(48 + \text{mystery}) // 64 \text{ +/- multiple of } 512 \text{ (number of cache sets)} = 436$
- ...

Suppose we determine that calling Check evicts from the data cache sets as follows:

key value | evicts values from cache set indexes

0	15, 72, 435
32	15, 72, 435
48	15, 72, 436
128	15, 72, 437
8240	15, 52, 72

Based on this information what is a possible value for mystery?

Answer:

Key: a value between 27856 and 27871 +/- any multiple of 32768

originally we erroneously wrote % 16384 instead of % 32768, and 4096 instead of 8240 for the last value

$(0 + \text{mystery}) // 64 \pm \text{multiple of } 512 \text{ (number of cache sets)} = 435$

$(32 + \text{mystery}) // 64 \pm \text{multiple of } 512 \text{ (number of cache sets)} = 435$

$(48 + \text{mystery}) // 64 \pm \text{multiple of } 512 \text{ (number of cache sets)} = 436$

...

let's choose the multiple of 512 to be 0 for simplicity, for now

$\text{mystery} // 64 = 435$ implies mystery in $[435 * 64 = 27840, 27840 + 63 = 27903]$

$(32 + \text{mystery}) // 64 = 435$ implies mystery in $[435 * 64 - 32 = 27808, 27808 + 63 = 27871]$

$(48 + \text{mystery}) // 64 = 436$ implies mystery in $[436 * 64 - 48 = 27856, 27856 + 63 = 27919]$

overlap here implies mystery in $[27856, 27871]$

the multiple of 512 offsets this by $512 * 64 = 32768$

extracting low-order bits

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
other_array[mystery * BLOCK_SIZE] += 1;
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

with 64KB direct-mapped cache with 64B blocks

suppose we find out that `array[0x700]` is slow to access

and `other_array` starts at some multiple of cache size

What was mystery?


```

char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array); // PRIME
other_array[mystery] += 1;
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) // PROBE
        {...}
}

```

- $NSETS = CACHE_SIZE / BLOCK_SIZE = 64KB / 64B = 1K = 2^{10}$
- And this affected `array[0x700]` //cache-aligned
 - Which had cache index $0x700 / BLOCK_SIZE = 1792 / 64 = 28$
 - Or `0b 0111 0000 0000`
- `other_array[mystery] = other_array + mystery` (because these are char array)
- If we know the base address of `other_array` is `0x20000`, we need `index(0x20000 + mystery) = 28`
- `0b 0010 0000 0000 0000 0000` //other_array
- `+0b 0000 0000 0000 0000 0000` //mystery
- `=0b 0010 0000 0000 0000 0000`
- Now we find the low order byte of mystery, which is `0b 0001 1100 = 28`
- In either case, we extract $\log(NSETS)$ bits, at the positions that align with the index bits

```

char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array); // PRIME
other_array[mystery] += 1;
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) // PROBE
        {...}
}

```

- $NSETS = CACHE_SIZE / BLOCK_SIZE = 64KB / 64B = 1K = 2^{10}$
- And this affected `array[0x700]`
 - Which had cache index $0x700 / BLOCK_SIZE = 1792 / 64 = 28$
 - Or `0b 0111 0000 0000`
- `other_array[mystery] = other_array + mystery` (because these are char array)
- If we know the base address of `other_array` is `0x20440`, we need to index $(0x20440 + mystery) = 28$
- `0b 0010 0000 0100 0100 0000 //other_array`
- `+0b 0000 0000 0010 11?? 0000 //mystery`
- `=0b 0010 0000 0110 11?? 0000`
- Now we find the actual value of `mystery`, which is `0b 0000 1011 = 11`

```

char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array); // PRIME
other_array[mystery * BLOCK_SIZE] += 1;
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) // PROBE
        {...}
}

```

- $\text{NSETS} = \text{CACHE_SIZE} / \text{BLOCK_SIZE} = 64\text{KB} / 64\text{B} = 1\text{K}$
- Each value of `mystery` touches a different cache line
 - So we touched cache index `mystery % NSETS`
 - But base address might be offset
- And this affected `array[0x700]`
 - Which had cache index $0x700 / \text{BLOCK_SIZE} = 1792 / 64 = 28$
- And `&other_array` starts at `0x20440`, which has cache index $(0x20440 / \text{BLOCK_SIZE}) \% \text{NSETS} = 17$
- So $\text{IDX}(\text{mystery}) + \text{IDX}(\&\text{other_array}) = 28$
- So $\text{IDX}(\text{mystery}) = 28 - 17 = 11$
- So `mystery = 11` or $(11 + 1024)$ or ...
 - If we know `mystery` is a char, then we know it's between 0-255, so in this case `mystery = 11`
- It's the same math!!!

```

char array[CACHE_SIZE] // not aligned
AccessAllOf(array); // PRIME
other_array[mystery * BLOCK_SIZE] += 1;
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) // PROBE
        {...}
}

```

- $NSETS = CACHE_SIZE / BLOCK_SIZE = 64KB / 64B = 1K$
- Each value of `mystery` touches a different cache line
 - So we touched cache index $mystery \% NSETS$
 - But base address might be offset
- And this affected `array[0x8280]`
 - Whose base address might also be offset, say 0x48480
 - What cache index is `array[0x8280]`?
 - $IDX(\&array + 0x8280) = ((0x48480 + 0x8280) / BLOCK_SIZE) \% NSETS = 28$
- And `&other_array` starts at 0x20440, which has cache index $(0x20440 / BLOCK_SIZE) \% NSETS = 17$
- So $IDX(mystery) + IDX(\&other_array) = 28$
- So $IDX(mystery) = 28 - 17 = 11$
- So `mystery = 11` or $(11 + 1024)$ or ...
 - If we know `mystery` is a `char`, then we know it's between 0-255, so in this case `mystery = 11`

What about associative caches?

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
other_array[mystery * BLOCK_SIZE] += 1;
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

with 64KB 2-way cache with 64B blocks

suppose we find out that `array[0x800]` is slow to access

and `other_array` starts at some multiple of cache size

What was mystery?

another exercise

```
char array1[...];  
...  
int secret;  
...  
y = array2[array1[x] * 4096];
```

- Suppose our cache has 64B blocks and 1K sets, and array2[0] is in set 0
- Suppose our prime+probe lets us see that something in cache set 256 or our probe array (array2) is evicted
- What do we know about array1[x]?

```
char array1[...];  
...  
int secret;  
...  
y = array2[array1[x] * 4096];
```

- Suppose our cache has 64B blocks and 1K sets, and array2[0] is in set 0
 - So array2[64] is in set 1, array2[128] is in set 2, etc.
- Suppose our prime+probe lets us see that something in cache set 256 of our probe array (array2) is evicted,
 - So $\text{CACHE_SET}(\text{array1}[x] * 4096) = 256$
- What do we know about array1[x]?
 - $\text{array1}[x] * 4K = 64 * \text{target_set} + \text{some multiple of number of sets}$
 - $\text{array1}[x] * 4K = 64 * 256 + \dots$
 - So $\text{array1}[x] = (64 * 256) / 4K = 16K / 4K = 4 + \dots$

```
char array1 [...];  
...  
int secret;  
...  
y = array2[array1[x] * 4096];
```

- Suppose our cache has 64B blocks and 32K sets, and array2[0] is in set 0
 - So array2[64] is in set 1, array2[128] is in set 2, etc.
- Suppose our prime+probe lets us see that something in cache set 256 of our probe array is evicted, so $\text{CACHE_SET}(\text{array1}[x] * 4096) = 256$
- What do we know about array1[x]?
 - $\text{array1}[x] * 4K = 64 * \text{target_set} + \text{some multiple of number of sets}$
 - $\text{array1}[x] * 4K = 64 * 256 + n * 32K * 64$
 - So $\text{array1}[x] = (64 * 256 + n * 32K * 64) / 4K = 16K / 4K + (n * 32K * 64) / 4K$
 - So array1[x] = 4 or 4+512 or...
 - But it's a char, so it can only be 4


```

char array1 [...];
...
int secret;
...
y = array2[array1[x] * 4096];

```

- Suppose our cache has 64B blocks and 2K sets, and array2[0] is in set 0
 - So array2[64] is in set 1, array2[128] is in set 2, etc.
- Suppose our prime+probe lets us see that something in cache set 256 of our probe array is evicted, so $\text{CACHE_SET}(\text{array1}[x] * 4096) = 256$
- What do we know about array1[x]?
 - $\text{array1}[x] * 4K = 64 * \text{target_set} + \text{some multiple of number of sets}$
 - $\text{array1}[x] * 4K = 64 * 256 + n * 2K * 64$
 - So $\text{array1}[x] = (64 * 256 + n * 2K * 64) / 4K = 16K / 4K + (n * 2K * 64) / 4K$
 - So array1[x] = 4 or 4+32 or 4+64 or...
 - But it's a char, so it can only be 4, 36, 68, 100, 132, 164, or 196
 - ... This works better in last-level caches with larger # of sets

```

char array1 [...];
...
int secret;
...
y = array2[array1[x]]; // no *4096 this time

```

- Suppose our cache has 64B blocks and 32K sets, and array2[0] is in set 0
 - So array2[64] is in set 1, array2[128] is in set 2, etc.
- Suppose our prime+probe lets us see that something in cache set 3 of our probe array is evicted, so $\text{CACHE_SET}(\text{array1}[x] \text{ ~~*4096~~)} = 3$
- What do we know about array1[x]?
 - $\text{array1}[x] \text{ ~~*4K~~} = 64 * \text{target_set} + \text{some multiple of number of sets}$
 - $\text{array1}[x] \text{ ~~*4K~~} = 64 * 3 + n * 32K * 64$
 - So $\text{array1}[x] = 196 + n * 32K * 64$
 - So $\text{array1}[x] = 196$ or some large number
 -

backup slides

beyond 1-bit predictor

devote *more space* to storing history

main goal: rare exceptions don't immediately change prediction

example: branch taken 99% of the time

1-bit predictor: wrong about 2% of the time

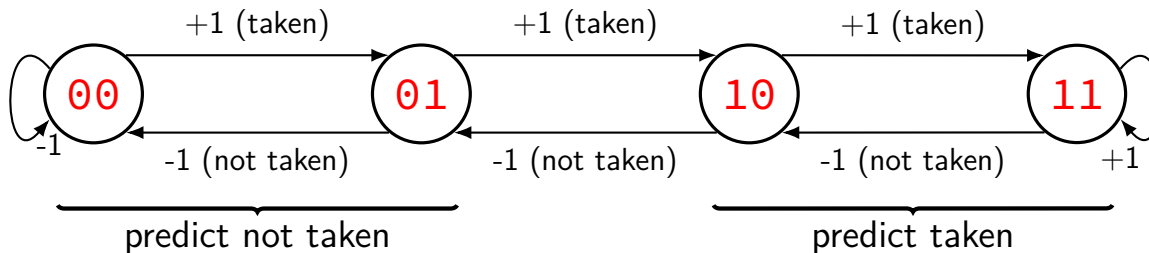
- 1% when branch not taken

- 1% of taken branches right after branch not taken

new predictor: wrong about 1% of the time

- 1% when branch not taken

2-bit saturating counter



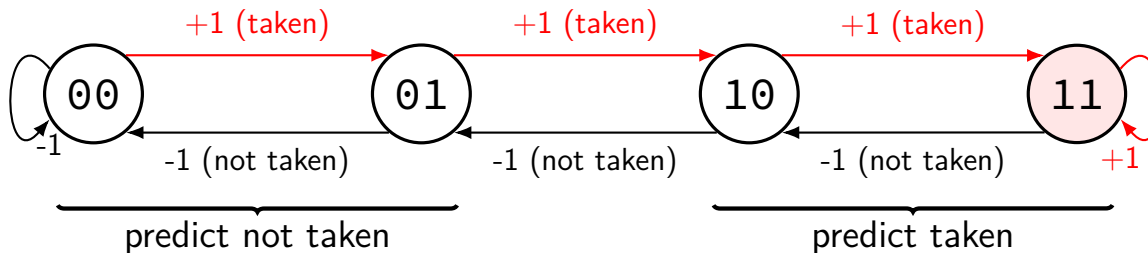
PC of branch

0x40042A

hash function

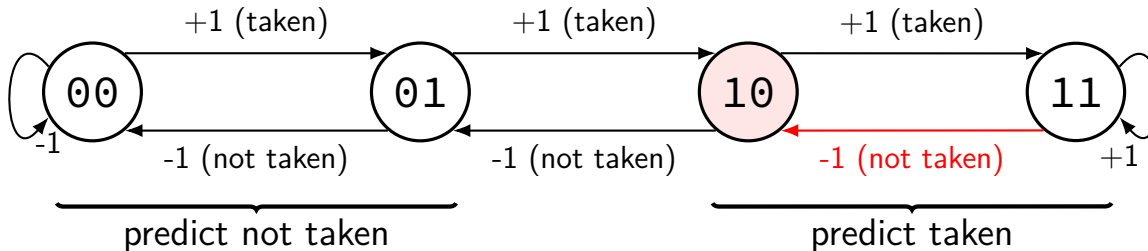
index	counter
0	11
1	01
2	11
...	...
14	10
15	00

2-bit saturating counter



branch always taken:
value increases to 'strongest' taken value

2-bit saturating counter



branch almost always taken, then not taken once:
still predicted as taken

example

0x40041B	movq \$4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

iter.	table before	prediction	outcome	table after
1	01	not taken	taken	10
2	10	taken	taken	11
3	11	taken	taken	11
4	11	taken	not taken	10
1	10	taken	taken	11
2	11	taken	taken	11
3	11	taken	taken	11
4	11	taken	not taken	10
1	10	taken	taken	11
...	...	...	...	...

generalizing saturating counters

2-bit counter: ignore one exception to taken/not taken

3-bit counter: ignore more exceptions

000 $\leftrightarrow$ 001 $\leftrightarrow$ 010 $\leftrightarrow$ 011 $\leftrightarrow$ 100 $\leftrightarrow$ 101 $\leftrightarrow$ 110 $\leftrightarrow$ 111

000-011: not taken

100-111: taken

exercise

use 2-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional branch misprediction rate?

```
int i = 0;
while (true) {
    if (i % 3 == 0) goto next;
    ...
next:
    i += 1;
    if (i == 50) break;
}
```

branch patterns

```
i = 4;  
do {  
    ...  
    i -= 1;  
} while (i != 0);
```

typical pattern for jump to top of do-while above:

TTTN TTTN TTTN TTTN TTTN...(T = taken, N = not taken)

goal: take advantage of recent pattern to make predictions

just saw 'NTTTNT'? predict T next

'TNTTTN'? predict T; 'TTNTTT'? predict N next

...

local pattern predictor (incomplete)

PC of branch

0x40042A

hash function

0x40041B	movq \$4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

4-iter loop: TTTN TTTN TTTN ...

idx recent
pattern

0	NNNNNN
1	NNTNTT
2	TTTTNT
3	TTTTTT
...	...
14	NTNTTN
15	NNTTTT

actual outcome

from commit(?) stage

???

convert to
prediction
???

prediction
to fetch stage

local pattern predictor (incomplete)

PC of branch

0x40042A

hash function

0x40041B	movq \$4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

idx recent
pattern

0	NNNNNN
1	NNTNTT
2	TTTTNT
3	TTTTTT
...	...
14	NTNTTN
15	NNTTTT

actual outcome

from commit(?) stage

???

convert to
prediction
???

prediction
to fetch stage

4-iter loop: TTTN TTTN TTTN ...

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT

local pattern predictor (incomplete)

PC of branch

0x40042A

hash function

0x40041B	movq \$4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

4-iter loop: TTTN TTTN TTTN ...

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT

idx recent
pattern

0	NNNNNN
1	NNTNTT
2	TTTNTT
3	TTTTTT
...	...
14	NTNTTN
15	NNTTTT

actual outcome
from commit(?) stage

???

convert to
prediction
???

prediction
to fetch stage

local pattern predictor (incomplete)

PC of branch

0x40042A

hash function

0x40041B	movq \$4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

idx recent
pattern

0	NNNNNN
1	NNTNTT
2	TTTNTT
3	TTTTTT
...	...
14	NTNTTN
15	NNTTTT

actual outcome

from commit(?) stage

???

convert to
prediction
???

prediction
to fetch stage

4-iter loop: TTTN TTTN TTTN ...

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT
2	TTTNTT	???	taken	TTNTTT

local pattern predictor (incomplete)

PC of branch

0x40042A

hash function

0x40041B	movq \$4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

idx recent
pattern

0	NNNNNN
1	NNTNTT
2	TTTNTTT
3	TTTTTT
...	...
14	NTNTTN
15	NNTTTT

actual outcome

from commit(?) stage

???

convert to
prediction
???

prediction
to fetch stage

4-iter loop: TTTN TTTN TTTN ...

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT
2	TTTNTT	???	taken	TTNTTT
3	TTNTTT	???	taken	TNTTTT

local pattern predictor (incomplete)

PC of branch

0x40042A

hash function

0x40041B	movq \$4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

idx recent
pattern

0	NNNNNN
1	NNTNTT
2	TTTTNTTT
3	TTTTTT
...	...
14	NTNTTN
15	NNTTTT

actual outcome

from commit(?) stage

???

convert to
prediction
???

prediction
to fetch stage

4-iter loop: TTTN TTTN TTTN ...

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT
2	TTTNTT	???	taken	TTNTTT
3	TTNTTT	???	taken	TNTTTT
4	TNTTTT	???	not taken	NTTTTN
1	NTTTTN	???	taken	TTTTNT
2	TTTTNT	???	taken	TTTNTT

recent pattern to prediction?

easy cases:

just saw TTTTTT: predict T

just saw NNNNNN: predict N

just saw TNTNTN: predict T

hard cases:

TTNTTTTT

predict T? loop with many iterations

(NTTTTTTTNTTTTTTTNTTTTTT...)

predict T? if statement mostly taken

(TTTTNTTNTTTTTTTTTTTNTTTT...)

predict N? loop with 5 iterations

(NTTTTNTTTTNTTTTNTTTTNTT...)

history of history

PC of branch

0x40042A

hash

idx	recent pattern
0	NNNN
1	TNTT
2	TTTN
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern

NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter

00
00
...
10
...
11
...
01
01
11

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT

prediction
to fetch sta

history of history

PC of branch

0x40042A

hash

idx	recent pattern
0	NNNN
1	TNTT
2	TTTN
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern

NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter

00
00
...
10
...
11
...
01
01 10
11

prediction
to fetch sta

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT

history of history

PC of branch

0x40042A

hash

idx	recent pattern
0	NNNN
1	TNTT
2	TTNT
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern
NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter
00
00
...
10
...
11
...
01
01 10
11

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT

prediction
to fetch sta

history of history

PC of branch

0x40042A

hash

idx	recent pattern
0	NNNN
1	TNTT
2	TTTNTT
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern

NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter

00
00
...
10
...
11
...
01 10
01 10
11

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT

prediction
to fetch sta

history of history

PC of branch

0x40042A

hash

idx	recent pattern
0	NNNN
1	TNTT
2	TTTNTT
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern
NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter
00
00
...
10
...
11
...
01 10
01 10
11

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT

prediction
to fetch sta

history of history

PC of branch

0x40042A

hash

idx	recent pattern
0	NNNN
1	TNTT
2	TTTNTT
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern

NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter

00
00
...
10
...
11
...
~~01~~ 10
~~01~~ 10
11

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT

prediction
to fetch sta

history of history

PC of branch

0x40042A

hash

idx	recent pattern
0	NNNN
1	TNTT
2	TTTNTTT
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern
NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter
00
00
...
10
...
11
...
01 10
01 10
11

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT

prediction
to fetch sta

history of history

PC of branch

0x40042A

hash

idx recent
pattern

0	NNNN
1	TNTT
2	TTTNTTT
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern

NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter

00
00
...
~~10~~ 11
...
11
...
~~01~~ 10
~~01~~ 10
11

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT

prediction
to fetch sta

history of history

PC of branch

0x40042A

hash

idx recent
pattern

0	NNNN
1	TNTT
2	TTTNTTT
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern

NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter

00
00
...
10 11
...
11
...
01 10
01 10
11

prediction
to fetch sta

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT
1	TTTN	10	taken	taken	11	TTNT

history of history

PC of branch

0x40042A

hash

idx recent
pattern

0	NNNN
1	TNTT
2	TTTNTTT
3	TTTT
...	...
14	NTTN
15	TTTT

actual outcome
from commit(?) stage

pattern

NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter

00
00
...
10 11
...
11
...
01 10
01 10 11
11

prediction
to fetch sta

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT
1	TTTN	10	taken	taken	11	TTNT

history of history

PC of branch

actual outcome
from commit(?) stage

0x40042A

hash

idx	recent pattern
0	NNNN
1	TNTT
2	TTTNTTTT
3	TTTT
...	...
14	TTNT
15	TTTT

pattern

NNNN
NNNT
...
NTTT
...
TNTT
...
TTNT
TTTN
TTTT

counter

00
00
...
10 11
...
11
...
01 10
01 10 11
11

iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT
1	TTTN	10	taken	taken	11	TTNT

prediction
to fetch sta

local patterns and collisions (1)

```
i = 10000;  
do {  
    p = malloc(...);  
    if (p == NULL) goto error; // BRANCH 1  
    ...  
} while (i-- != 0); // BRANCH 2
```

what if branch 1 and branch 2 hash to same table entry?

local patterns and collisions (1)

```
i = 10000;  
do {  
    p = malloc(...);  
    if (p == NULL) goto error; // BRANCH 1  
    ...  
} while (i-- != 0); // BRANCH 2
```

what if branch 1 and branch 2 hash to same table entry?

pattern: TNTNTNTNTNTNTNTNT...

actually no problem to predict!

local patterns and collisions (2)

```
i = 10000;  
do {  
    if (i % 2 == 0) goto skip; // BRANCH 1  
    ...  
    p = malloc(...);  
    if (p == NULL) goto error; // BRANCH 2  
skip: ...  
} while (i-- != 0); // BRANCH 3
```

what if branch 1 and branch 2 and branch 3 hash to same table entry?

local patterns and collisions (2)

```
i = 10000;  
do {  
    if (i % 2 == 0) goto skip; // BRANCH 1  
    ...  
    p = malloc(...);  
    if (p == NULL) goto error; // BRANCH 2  
skip: ...  
} while (i-- != 0); // BRANCH 3
```

what if branch 1 and branch 2 and branch 3 hash to same table entry?

pattern: TTNNTTNNTTNNTTNNTT

also no problem to predict!

local patterns and collisions (3)

```
i = 10000;  
do {  
    if (A) goto one  // BRANCH 1  
    ...  
one:  
    if (B) goto two  // BRANCH 2  
    ...  
two:  
    if (A or B) goto three // BRANCH 3  
    ...  
    if (A and B) goto three // BRANCH 4  
    ...  
three:  
    ... // changes A, B  
} while (i-- != 0);
```

what if branch 1-4 hash to same table entry?

better for prediction of branch 3 and 4

global history predictor: idea

one predictor idea: ignore the PC

just record taken/not-taken pattern for all branches

lookup in big table like for local patterns

global history predictor (1)

branch history register

NTTT

pat

NNNN

NNNT

...

NTTT

TNNN

TNNT

TNTN

...

TTTN

TTTT

counter

00

00

...

10

01

10

11

...

10

11

outcome
from
commit(?)

prediction
to fetch stage

global history predictor (1)

```

i = 10000;
do {
    if (i % 2 == 0) goto skip;
    ...
    if (p == NULL) goto error;
skip:
    ...
} while (i-- != 0);
    
```

branch history register

NTTT

pat

NNNN

NNNT

...

NTTT

TNNN

TNNT

TNTN

...

TTTN

TTTT

counter

00

00

...

10

01

10

11

...

10

11

outcome
from
commit(?)

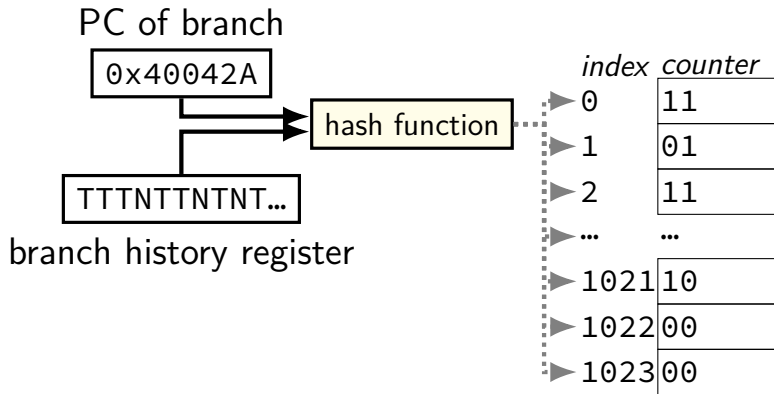
prediction
to fetch stage

iter./ branch	history before	counter before	predict	outcome	counter after	history after
0/mod 2	NTTT	10	taken	taken	11	TTTT
0/loop	TTTT			taken		TTTT
1/mod 2	TTTT			not taken		TTTN
1/error	TTTN			not taken		TTNN
1/loop	TNNT			taken		NNTT
2/mod 2	NNTT			taken		NTTT
2/loop	TTTT			taken		TTTT

correlating predictor

global history *and* local info good together

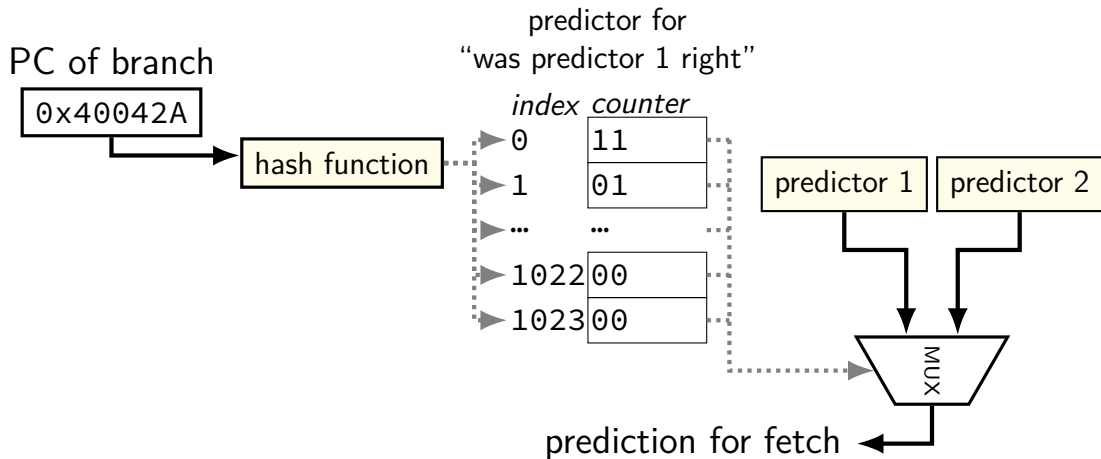
one idea: **combine history register + PC** (“gshare”)



mixing predictors

different predictors good at different times

one idea: have two predictors, + predictor to predict which is right



loop count predictors (1)

```
for (int i = 0; i < 64; ++i)  
    ...
```

can we predict this perfectly with predictors we've seen

yes — local or global history with 64 entries

but this is very important — more efficient way?

loop count predictors (2)

loop count predictor idea: look for NNNNNNT+repeat (or TTTTTTN+repeat)

track for each possible loop branch:

- how many repeated Ns (or Ts) so far

- how many repeated Ns (or Ts) last time before one T (or N)

- something to indicate this pattern is useful?

known to be used on Intel

benchmark results

from 1993 paper

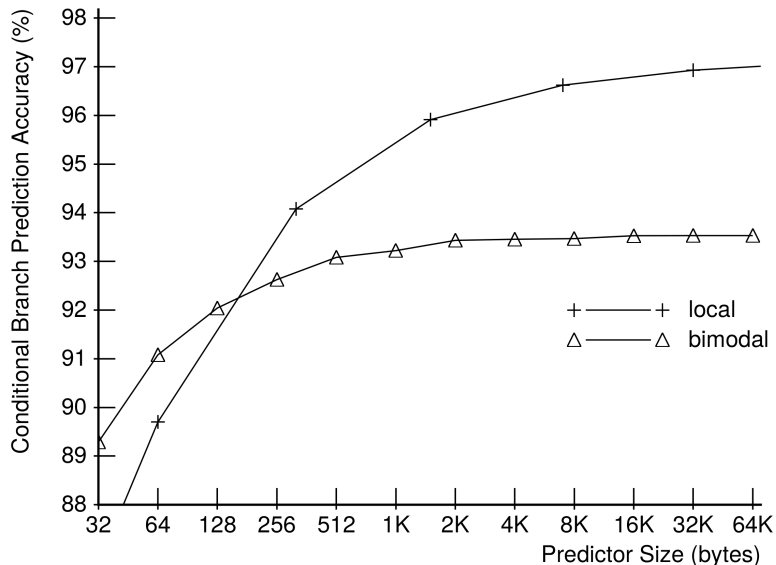
(not representative of modern workloads?)

rate for conditional branches on benchmark

variable table sizes

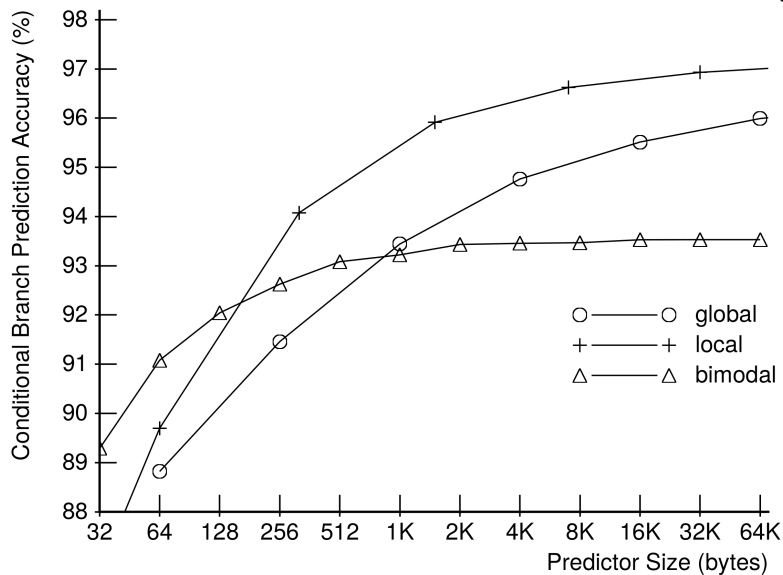
2-bit ctr + local history

from McFarling, "Combining Branch Predictors" (1993)



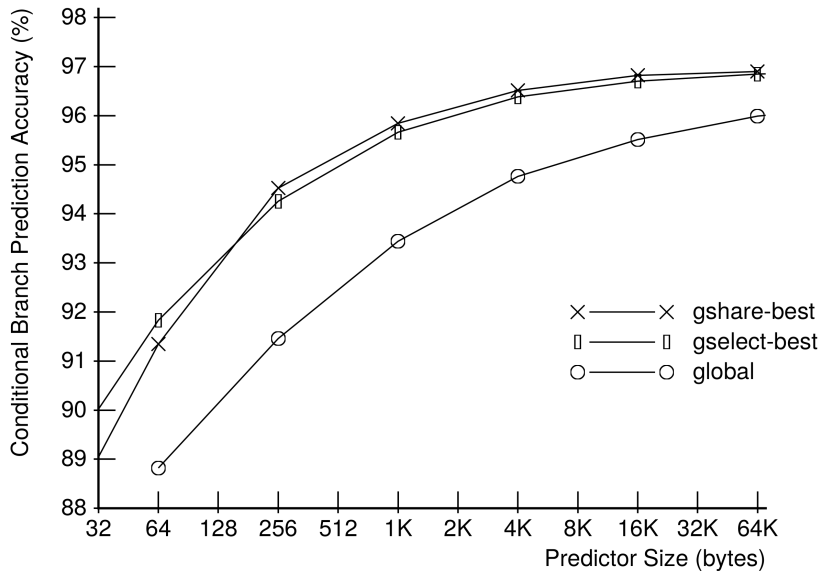
2-bit (bimodal) + local + global hist

from McFarling, "Combining Branch Predictors" (1993)



global + hash(global+PC) (gshare/gselect)

from McFarling, "Combining Branch Predictors" (1993)



real BP?

details of modern CPU's branch predictors often not public
but...

Google Project Zero blog post with reverse engineered details

`https://googleprojectzero.blogspot.com/2018/01/reading-privileged-memory-with-side.html`

for RE'd BTB size:

`https://xania.org/201602/haswell-and-ivy-btb`

reverse engineering Haswell BPs

branch target buffer

- 4-way, 4096 entries

- ignores bottom 4 bits of PC?

- hashes PC to index by shifting + XOR

- seems to store 32 bit offset from PC (not all 48+ bits of virtual addr)

indirect branch predictor

- like the global history + PC predictor we showed, but...

- uses history of recent branch addresses instead of taken/not taken

- keeps some info about last 29 branches

what about conditional branches??? loops???

- couldn't find a reasonable source